Random variables and their distributions 3 unit

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Review of Probability

Introduction

The theory of probability is one of the most useful and interesting branches of modern mathematics. It is becoming prominent by its applications in many fields of learning, such as insurance, statistics, space science, Engineering, etc.

An Italian mathematician, Galilio (1564-1642) attempted a quantitative measure of probability while dealing some problems related to gambling. In the middle 17th Century Two French mathematicians, Pascal and Fermat laid down the first foundation of Mathematical theory of probability while solving the famous "problem of points" posed by Chevalier-De-mere.

Statistics, is asenee, is a old as the human society itself. In India an efficient System of collecting official and administrative statistics existed even were than 2000 year ago, in particular, during the reign of Chandra Gupta Maurya (324-300 B.C). From Kautilya's Arthashastra it is known that even before 300 B.C. a very good System of collecting "Vital Statistics" and registation of births and deaths was in vogue.

During Akbar's reign (1556-1605) A.D. Raja Todalmal, the then Land and revenue minister, maintained good records ~~in~~ of Land and agricultural statistics. In Aina-e Akbari written by Abul Fazal, one of the nine gems of Akbar, we find detailed accounts of the administrative and statistical Surveys conducted by Akbar's reign.

Random Experiment

If an 'experiment' is conducted any number of times, under essentially identical conditions, there is a set of all possible out comes, the experiment is called a random trial or random experiment.

The out comes are known as elementary events and a set of out comes is an event. Thus an elementary event ~~is~~ is also an event.

Sample space:

The set of all possible out comes of a random experiment is called sample space, it is denoted by S.

The elements of sample space are sample points.

Ex: In the toss of a single coin let the out come "Head turning up" be denoted by H and the out come "Tail turning up" be denoted by T

The coin is repeatedly tossed under the same conditions. Then the toss of the coin results in the out come H or T. thus yielding the sample space $S = \{H, T\}$.

Finite Sample Space: A sample space is called finite Sample space, if its sample points are finite number.

Infinite Sample Space: A sample space is called infinite Sample space, if its sample points are infinite numbers.

A bulb is allowed to burn continuosly till it expires. Then the sample point (out come) e may be expressed in number of hours (minuts and seconds ignored) and $e > 0$ then the sample space $S = \{1, 2, \ldots\}$ which is an infinite ~~sample space is~~ set. implies that it is an infinite sample space.

In order to model analyze a random experiment, we must understand the collection of possible out comes of the experiment. For this, in this introduction to probability, we use the basic concepts of sets and operations on sets, because a sample space or an experiment is the set s of possible out comes of the experiment.

Any subset E of a sample space S is called event.

Ex. 1. Tossing of a coin is a random experiment, Here the Sample space $S = \{H, T\}$, also $\{H\}, \{T\}, \{H, T\}, \{\}$ are events.

The event S is called the sure event or certain event. and the event ϕ is called an impossible event.

Exhaustive Events.

Probability:

The total no. of possible out comes individually in any random experiment is known as exhaustive event.

Ex. 1. In tossing a coin, there are two exhaustive events they are $\{H\}, \{T\}$, Since $S = \{H, T\}$.

Ily tossing two coins we have

$S = \{HH, HT, TT, TH\}$, Therefore the no. of exhaustive events are 4 they are

$\{HH\}\ \{HT\}\ \{TT\}\ \{TH\}$.

Clearly, the total no. of exhausting events when tossing n coins is given by 2^n.

Ex. 2. In throwing a single die, we have $S = \{1,2,3,4,5,6\}$ Here, the no. of exhaustive events are 6 they are

$\{1\}\ \{2\}\ \{3\}\ \{4\}\ \{5\}\ \{6\}$.

Ily when we throwing two dice, the total no. of exhaustive events are 36. Sime here

$S = \{(1,1)(1,2)\ldots(1,6), (2,1)(2,2)\ldots(2,6)$
$(3,1)(3,2)\ldots(3,6), (4,1)(4,2)\ldots(4,6)$
$(5,1)(5,2)\ldots(5,6)\quad (6,1)(6,2)\ldots(6,6)\}$.

Clearly, when we are throwing n dice the total no. of exhaustive events are 6^n.

Mutually Exclusive Events. (disjoint events).

The events are said to be mutually exclusive if the happening of any one event ruled out happening of all other events.

i.e., if no two or more than two events can happen simultaneously in same random experiment.

In other words, two events E_1 and E_2 said to be mutually exclusive events or disjoint if $E_1 \cap E_2 = \{\}$

Example

Consider a random experiment as throwing a die Here $S = \{1,2,3,4,5,6\}$. Let $E_1 = \{2,4,6\}$ $E_2 \{1,3,5\}$.

$E_3 \{3,6\}$.

Clearly E_1 and E_2 are mutually exclusive events and E_1 and E_3 are not a mutually exclusive events.

$$E_1 \cap E_3 = \{6\} \neq \{\}.$$

Equally likely events:

Events are said to be equally likely if there is a no reason to expect any one in presence to other.

Ex: In throwing a die, all the 6 faces are equally likely to come out therefore $\{1\}\{2\}\{3\}$ $\{4\}\{5\}\{6\}$. all are equally likely events.

Independent and Dependent Events.

Two or more events are said to be independent if the happening or not happening of any one event does not depend by the happening or not happening of any other event. Otherwise, they are said to be dependent.

Suppose a box contains 4 black balls and 3 white balls and let E_1 be the event that first ball drawn is black and let E_2 be the event the second ball drawn is black then

(i) the balls are not replaced after being drawn then E_1 and E_2 are dependent events.

(ii) the ball are replaced after being drawn then E_1 and E_2 are independent.

Probability: In a random experiment let there be n exhaustive mutually exclusive equally likely events, and m of them are favourable to the happening of an event 'E', then the probability happening of E is given by

$$P(E) = \frac{\text{no. of favourable out comes for } E}{\text{total no. of out comes.}}$$

$$\therefore P(E) = m/n.$$

Observation: $P(\bar{E}) = \frac{\text{no. of favourable out comes for } \bar{E}}{\text{total no. of out comes.}}$

$$= \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(E).$$

$$\therefore \boxed{P(\bar{E}) = 1 - P(E)}.$$

1. If E is an event, then $0 \leq P(E) \leq 1$.

2. If S is a the sample space of any random experiment, then $P(S) = 1$.

3. If A and B two mutually exclusive events then $P(A \cup B) = P(A) + P(B)$

Illy $E_1, E_2 \ldots E_n$ are mutually exclusive events

$P(E_1) + P(E_2) + \ldots P(E_n) = P(E_1 \cup E_2 \cup \ldots E_n)$

Random Variables and their distributions

Previously we considered the concept of random experiment, events, sample space and sample points. the events described on a random experiment may be numerical or non-numerical (descriptive).

For example, the out comes that we obtain, when we throw a die are numerical. (We get out comes as 1,2,3,4,5 or 6) Hence the sample space is numerical. But the out comes we obtain, when we toss a coin are non-numerical. We get the out comes as a head or a tail.

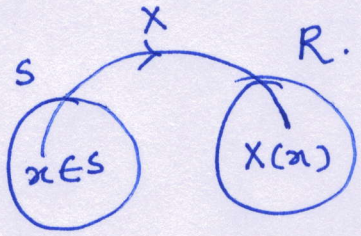
Here the sample-space is non-numerical or discriptive. It is inconvenient to deal with these non-numerical descriptive out comes mathematically. Hence, for ease of manipulation, we may assign a real number to each of the out comes using a fixed rule or mapping.

For example, when we toss a coin we get two out comes, namely Head or tail. We can assign a numerical values say 1 for Head and 0 for tail. This interpretation is easy and attractive from mathematical point of view and also practically meaningfull.

This rule or mapping from the orginal sample space (numerical or non-numerical) to a numerical (real) sample space, subject to certain constraints is called a random variable.

In general random variable means a real number associated with out comes of a random experiment.

A random variable X can also defined by a real Valued function on the sample space S of a random experiment such that for each point x of the Sample space, $X(x)$ is the probability of occurence of the event represented by x.



Example: The sample space corresponding to tossing of two coins.

When two coins are tossed, its out comes or Sample points can be (Head, Head) (H,T) (Tail H) (T,T). i.e., $S = \{HH, HT, TH, TT\}$

After the performance of the experiment, we count the number of tails and denote it by X.

First out come HH has 0 tail. So X=0, Simly X=1. denote the out comes HT or TH and X=2, represents the out comes. TT.

Thus X takes the values 0, 1, 2.

i.e., X = 0, 1, 2.

A random variable is also called a stochastic variable or simple ~~points~~ variable or a chance variable. Random variables are usually denoted by capital letters of English alphabets and particular values which the random variable takes are denoted by the corresponding small letters.

A Random variable is any function that assigns a numerical value to each possible outcome.

Ex-1.

Ex. 1. Consider a random experiment consisting of tossing a coin twice. The sample space $S = \{s_1, s_2, s_3, s_4\}$ where s_1 HH, s_2 = HT s_3 = TH s_4 = TT consists of four elements (sample points).

Define a function $X : S \to R$ by $X(s)$ = number of heads then $X(s_1) = 2$, $X(s_2) = 1$ $X(s_3) = 1$ $X(s_4) = 0$.

The Range of $X = \{x(s) : s \in S\}$.

$= \{X(s_1), X(s_2), X(s_3), X(s_4)\}$.

$= \{0, 1, 2\}$

Ex 2. Consider the random experiment of throwing a pair of dice and nothing the sum. The sample space consists of points $S = \{(1,1)(1,2)..(1,6)(2,1)(2,2)\cdots(2,6) \cdots (6,1)(6,2)\cdots(6,6)\}$.

If X is the "sum of number appearing on the faces of dice", X is a random variable.

Here $X : S \to R$ is defined as $X(s) = X((i,j)) = i+j$ for $(i,j) \in S$.

If in a random experiment, the event corresponding to a number a occurs, then the corresponding random variable X is said to assume the value a and the probability of the event is denoted by $P(X=a)$. Similarly, the probability of the event X assuming any value between a and b is denoted by $P(a<X<b)$. The probability of the event $x \leq c$ (X assumes any value smaller than or equal to c). is denoted by $P(X \leq c)$ and the probability of the event $X>c$ is denoted by $P(X>c)$. That is, $P(X>c)$ is simply the probability of the set of outcomes s for which $X(s)>c$ or $P(X>c) = P\{s \mid X(s)>c\}$.

for the first example $P(X \leq 1) = P\{HH, HT, TH\}$

$= 3/4$.

Types of Random Variables.

Random Variable is of two types.

1. Discrete Random Variable
2. Continuous Random Variable.

1. Discrete Random Variable: A random Variable X which can take only finite number of descrete values in an interval of domain is called Discrete Random Variable.

In otherwords, if the random variable takes the values only on the set $\{0, 1, 2 \ldots n\}$ is called Discrete Random Variable.

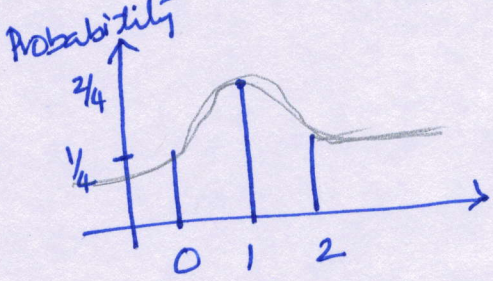
Ex. Tossing a coin, Throwing a dice, the number of defectives in a sample of electric bulbs, The number of printing mistakes in each page of a book, the number of Telephone calls received by the telephone operator are examples of Discrete random variable. ask example

2. Continuous Random Variable: A random Variable X which take values continuously, i.e, which take all possible values in a given interval is called a continuous random variable.

Suppose we want to know the probability that if an accident occurs on a free way whose length is 200 miles, it will happen at some given location or, perhaps, some particular stretch of the road. The out comes of this experiment can be looked upon as a continuous points. Namely, those on the continuous interval from 0 to 200.

Distributions of Probability over number of Heads.

When two coins are tossed, the number of Heads is a random Variable which can take the values 0, 1, 2. The associated Probabilities are $\frac{1}{4}$, $\frac{1}{2}$ & $\frac{1}{4}$ and is shown below.



It may be noted that total of all the three probabilities $\frac{1}{4}$, $\frac{1}{2}$ & $\frac{1}{4}$ is 1. This is so because the random variable is bound to take one three values 0, 1, or 2 in its range.

The above diagram shows how the total probability (=1) is distributed over the range of the variable.

So these diagrams are referred to graphs of probability distributions.

In many situations, these graphs are described by a mathematical function which is referred to as a Probability Function or statistical distribution.

Probability function of a Discrete Random Variable.

If for a desssete random variable X, the real valued function $P(x)$ is such that $P(X=x) = P(x)$. then $P(x)$ is called probability density function of a descrete random variable X. The probability function $P(x)$ gives the measure of probability for different values of x.

Properties of probability function :

If $P(x)$ is a probability function of a random variable X. then it posses the following properties.

1. $P(x) \geq 0$ for all x
2. $\sum P(x) = 1$. Summation is taken over for all values of y
3. $P(x)$ cannot be nagative for any value of x.

Probability Distribution Function :

Let X be a random variable, then the probability distribution function associated with X is defined as the probability that the out come of an experiment will be one of the out comes for which $X(s) \leq x \quad x \in R$. That is the function $F(x)$ [i.e., $F_x(x)$] defind by

$$F_x(x) = P(X \leq x) = P\{s : X(s) \leq x\}. \quad -\infty < x < \infty$$

is called the distribution fn. of X.

Discrete Probability Distribution (Probability Mass Function).

Probability distribution of a random variable is the set of its possible values together with their respective probabilities.

In other words Any function $f(x)$ satisfies above 3 conditions will automatically be a discrete probability mass function simply discrete probability function.

Suppose X is a discrete random variable with possible out comes $x_1, x_2 \ldots x_n$ then the probability of the each possible out come x_i is $P(x_i) = P(X = x_i)$.

If $P(x_i)$, $i = 1, 2, \ldots$ satisfy the two conditions

(i) $P(x_i) > 0$. for all values of $i = 0, 1, 2 \ldots$

(ii) $\sum P(x_i) = 1$, $i = 1, 2, 3 \ldots$ then the function $P(x)$ is called the probability mass function of the random variable X and the set $\{P(x_i)\}$ $i = 1, 2, \ldots$ is called the discrete random variable X.

The probability distribution of the random variable X is given by the following table.

X	x_1	x_2	x_3	$\ldots$	x_n
$P(x)$	$P(x_1)$	$P(x_2)$	$P(x_3)$	$\ldots$	$P(x_n)$.

Further $P(X < x_i) = P(x_1) + P(x_2) + \cdots P(x_{i-1})$

$P(X \leq x_i) = P(x_i) + P(x_2) + \cdots P(x_{i-1}) + P(x_i)$

and $P(x > x_i) = 1 - P(X \leq x_i)$.

Ex. 1. In tossing a coin two times, $S = \{TT, HH, HT, TH\}$.

focusson Head.

Random variable getting a Head.

$P(X=0)$ = Probability of getting no heads (TwoTails).

$= P(\{T,T\}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

$P(X=1)$ = Probability of getting one Head.

$= P\{\{HT, TH\}\} = \frac{2}{4} = \frac{1}{2}$.

$P(X=2)$ = Probability of getting two Heads.

$= P(\{H,H\}) = \frac{1}{4}$.

The total probability 1 is distributed into three points $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ according to whether $X = 0$ or 1 or 2. This probability distribution is given in the following table.

$X = x_i$	0	1	2
$P(X = x_i)$	1/4	1/2	1/4

Ex. Check whether the following can serve as (discrete) probability function

(a) $f(x) = \frac{x-2}{2}$ for $x = 1, 2, 3, 4$

(b) $h(x) = x^2/25$ for $x = 0, 1, 2, 3, 4$

(a) this function cannot serve as probability distribution because $f(1)$ is $-ve$

(b) this function cannot serve as probability distribution because the sum of the five probabilities is $6/5$ is not equal to 1.

Cummulative Distribution Function: $F(x)$ of a random variable X is defined by $F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$

Where $f(x)$ is the discrete probability function

Clearly, by the above definition, $F(x)$ can be obtains from $f(x)$ as follows.

$$F(x) = \begin{cases} 0 & -\infty < x < x_1 \\ f(x_1) & x_1 \leq x \leq x_2 \\ f(x_1)+f(x_2) & x_2 \leq x \leq x_3 \\ \vdots & \\ f(x_1)+\cdots f(x_n) & x_n \leq x < \infty \end{cases}$$

Where $x_1 x_2 \cdots x_n$ are the values of the discrete random variable X.

Ex: The descrete probality function $f(x)$ of X of the random experiment of 3 independent tosses of the fair coin.

$X = x_i$	0	1	2	3
$f(x_i) = P(X = x_i)$	$\frac{1}{8}$	$3/8$	$3/8$	$1/8$.

Then the Cummulative Probability function of X is given by

$$F(x): \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{8} & \text{if } 0 \le x < 1 \\ \frac{1}{2} & \text{if } 1 \le x < 2 \\ \frac{7}{8} & \text{if } 2 \le x < 3. \\ 1 & \text{if } 3 \le x < \infty \end{cases}$$

Conversly $f(x)$ can be obtained from $F(x)$. The following example shows how $F(x)$ can be used to determine $f(x)$ of a discrete random variable.

Suppose the cummulative distribution function of d.r.v X is

$$F(x) = \begin{cases} 0 & x < -2 \\ 0.2 & -2 \le x < 0 \\ 0.7 & 0 \le x < 2 \\ 1 & 2 \le x. \end{cases}$$

Determine the probability mass function $f(x)$ of d.r.v X.

From the definition of $F(x)$

$$f(x_1) = f(-2) = 0.2$$

$$f(x_1) + f(x_2) = f(-2) + f(0) = 0.7.$$

$$\Rightarrow f(0) = 0.7 - f(-2) = 0.7 - 0.2$$

$$= 0.5.$$

$$f(x_1) + f(x_2) + f(x_3) = 1$$

$$f(-2) + f(0) + f(2) = 1.$$

$$0.2 + 0.5 + f(2) = 1 \Rightarrow f(2) = 0.3.$$

x	-2	0	2
$f(x)$	0.2	0.5	0.3

Probability Density function

The probability density function $f(x)$ is defined as the derivative of probability $f_x(x) = \frac{d}{dx}[F_x(x)]$

Expectation of Discrete Variable:

Suppose a random variable X assumes the values $x_1, x_2 \ldots x_n$ with respective probabilities $P_1, P_2 \ldots P_n$. Then the mathematical Expectation or Mean or Expected Value of X, denoted by $E(X)$, is defined as the sum of products of different values of x and the corresponding probabilities

$$\therefore E(X) = x_1P_1 + x_2P_2 + x_3P_3 + \cdots + x_nP_n.$$

$$\text{i.e., } E(X) = \sum_{i=1}^{n} P_i x_i$$

$E(X) = \mu$ is called mean.

If X is a random variable and k is the constant, then $E(X+K) = E(X) + k$.

$$E(X) = \sum_{n=1}^{n} x_i P(x_i) = \sum_{n=1}^{n} x_i P_i$$

$$E(X+K) = \sum_{n=1}^{n} (x_i + k) P(x_i) = \sum_{i=1}^{n} x_i p_i + K \sum_{i=1}^{n} P_i$$

$$= E(X) + K(1)$$

$$E(X+K) = E(X) + 1.$$

(ii) If X and Y are two random variables, then

$E(X+Y) = E(X) + E(Y)$ provided that

$E(X)$ & $E(Y)$ Exist.

Let X assume the values $x_1, x_2 \cdots x_n$ and Y assumes the values $y_1, y_2 \cdots y_n$

Then by the definition,

$$E(X) = \sum_{i=1}^{n} p_i x_i \quad \text{and} \quad E(Y) = \sum_{i=1}^{n} p_i y_i$$

Let $p_{ij} = P(X = x_i \cap Y = y_j) = P(x_i\, y_i)$.

This called joint probabilityfn of X and Y.

The Sum $(X+Y)$ is also a random variable which can take $m \times n$ values $(x_i + y_j)$, $i = 1, 2 \cdots n$ and $j = 1, 2, \cdots m$.

$\therefore$ By definition

$$E(X+Y) = \sum_{i=1}^{n}\sum_{i=1}^{m} p_{ij}(x_i + y_j) = \sum_{i=1}^{n}\sum_{j=1}^{m} p_{ij} x_i + \sum_{i=1}^{n}\sum_{j=1}^{m} p_{ij} y_j$$

$$= \sum_{i=1}^{n}\left[x_i \sum_{j=1}^{m} p_{ij}\right] + \sum_{j=1}^{m}\left[y_j \sum_{i=1}^{n} p_{ij}\right].$$

$$= \sum_{i=1}^{n} x_i p_i + \sum_{j=1}^{m} y_j p_j = E(X) + E(Y).$$

By $E(X_1 + X_2 \cdots X_n) = E(X_1) + E(X_2) + \cdots E(X_n)$.

Note: 1. Mean $E(x) = \mu = \sum_{i=1}^{n} x_i p_i$

2. Variance, $Var(X) = E(X^2) - (E(x))^2$.

3. Standard Deviation $S.D = \sigma = \sqrt{\text{variance}}$.

Prob: A random variable X has the following probability function

x	0	1	2	3	4	5	6	7
P(x)	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

(i) Determine k (ii) Evaluate $P(X<6)$, $P(X \geq 6)$, $P(0<x<5)$ and $P(0 \leq X \leq 4)$.

(iii) if $P(X \leq K) >$

(iv) Determine the Distribution function of X, (v) Mean (vi) Variance

i) Since $\sum_{x=0}^{7} P(x) = 1$, we have

$$0+K+2K+2K+3K+K^2+2K^2+7K^2+K=1.$$

$$10K^2+9K-1=0.$$

$$(10K-1)(K+1)=0.$$

$K = \frac{1}{10} \neq 0-1$ sin $P(x) \geq 0$). So $K \neq -1$.

(ii) $P(X<6) = P(X=0) + P(X=1) + \cdots P(X=5)$

$= 0 + K + 2K + 2K + 3K + K^2$

$= 8K + K^2 = 8(0.1) + (0.1)^2$ $\because K = 0.1$

$= 0.8 + 0.01$

$= 0.81$

$P(X \geq 6) = 1 - P(X \leq 6)$

$= 1 - 0.81 = 0.19$.

or $P(X \geq 6) = 2k^2 + 7k^2 + k$.

$= 9k^2 + k = 9(0.1)^2 + 0.1$

$= 9(0.01) + 0.1$

$= 0.09 + 0.1 = 0.19$.

$P(0 < X < 5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$.

$= k + 2k + 2k + 3k = 8k$.

$= 8 \cdot \frac{1}{10} = \frac{8}{10} = 0.8$

$P(0 \leq X \leq 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$+ P(X=4)$.

$= 0 + k + 2k + 2k + 3k$.

$= 8k = 8/10 = \underline{\underline{0.8}}$.

The distribution function of X is given by the following

X	$P(x) = P(X \leq x)$.
	$= 0$
0	$k = 1/10$
1	$3k = \frac{3}{10}$
2	$5k = 5/10$
3	$8k = 8/10$
4	$8k + k^2 = 81/100 = 0.81$
5	$8k + 3k^2 = 83/100$
6	$9k + 10k^2 = 1$.
7	

Mean $\mu = \sum_{i=0}^{7} P_i x_i$

$= 0(0) + 1(K) + 2(2K) + 3(2K) + 4(3K)$
$+5(K^2) + 6(2K^2) + 7(7K^2+K)$.

$= 66K^2 + 30K = \frac{66}{100} + \frac{30}{10} = 0.66 + 3.00$

$= 3.66$ $\quad \because K = \frac{1}{10} = 0.1$.

Variance : $\sigma^2 = \sum_{i=0}^{7} P_i x_i^2 - \mu^2$

$= 0 \cdot 0^2 + K \cdot 1^2 + 2K \cdot 2^2 + 2K \cdot 3^2 + 3K \cdot 4^2$
$K^2 \cdot 5^2 + 2K^2 \cdot 6^2 + (7K^2+K) \cdot 49 - (3.66)^2$

$= 440K^2 + 124K - (3.66)^2 = \frac{440}{100} + \frac{124}{10} - (3.66)^2$

$= 4.4 + 12.4 - 13.3956$

$= 3.4044$.

Prob: A player tosses two fair coins. He wins Rs 100/- if Head appears, Rs 200/- if two Heads appear. On the other hand he loses Rs 500/- if no. head appears. Determine the expected Value E of the game and and is the game favourable to the player.

Let X denote the number of heads occuring in tosses of two fair coins.

The sample space is $S = \{HH, HT, TH, TT\}$

ie., $\{H,T\} \times \{H,T\}$.

Probability of 2 Heads all Heads $= P(X=2) = \frac{1}{4}$

Probability of 0 Heads " " $= P(X=0) = 1/4$
(i.e, 2 tails)

Probability of one Head. $P(X=1) = \frac{2}{4} =$

Discrete Probability Distribution is.

$X = x_i$	0	1	2
$P(x_i)$	1/4	2/4	1/4.

Expected Value of the game $E(x) = \mu =$ mean.

$$= 100 \times \frac{2}{4} + 200 \times \frac{1}{4} - 500 \times \frac{1}{4}.$$

$$= \frac{200 + 200 - 500}{4} = -\frac{100}{4} = -25 \text{ Rs.}$$

$\therefore$ Game is not favourable to the player since $E < 0$.

Prob: A sample of 4 items is selected at random from a box containing 12 items of which 5 are defective. Find the expected number of E of defective items.

Sol: let X be denote the numbers of defective items among 4 items drawn from 12 items.

obviously X can take the values 0, 1, 2, 3 or 4.

NO. of good items. 7.

NO. of defective items. 5.

$P(X=0) = P(\text{no defective}) = \frac{^7C_4}{^{12}C_4} = \frac{^7C_4}{495} = \frac{35}{495} = \frac{7}{99}$

$P(X=1) = P(\text{one defective item and 3 good items}) = \frac{^7C_3 \times {}^5C_1}{^{12}C_4} = \frac{7.6.5.5}{6}$

$= \frac{35}{99}$.

$P(X=2) = P(\text{two defective items and 2 not defective})$.

$= \frac{^7C_2 \times {}^5C_2}{^{12}C_4} = \frac{42}{99}$.

$P(X=3) = P(3 \text{ defective one not defective})$.

$= \frac{^5C_3 \times {}^7C_1}{^{12}C_4} = \frac{14}{99}$.

$P(X=4) = P(4 \text{ defective items})$. all are defective

$= \frac{^5C_4}{^{12}C_4} = \frac{5}{495} = \frac{1}{99}$.

Discrete probability distribution.

$X = x_i$	0	1	2	3	4
$P(X=x_i) = f(x_i)$	$\frac{7}{99}$	$\frac{35}{99}$	$\frac{42}{99}$	$\frac{14}{99}$	$\frac{1}{99}$.

Expected number of defective items

$$= E(X) = \sum x_i f_i$$

$$= 0 \cdot \frac{7}{99} + 1 \cdot \frac{35}{99} + 2 \cdot \frac{42}{99} + 3 \cdot \frac{14}{99} + 4 \cdot \frac{1}{99}.$$

$$= \frac{165}{99}.$$

Prob: The random variable X has probability function of the following form

$$f(x) = \begin{cases} K & \text{if } x = 0 \\ 2K & \text{if } x = 1 \\ 3K & \text{if } x = 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) Determine the value of K (b) find $P(X<2), P(X \leq 2)$ $P(0<X<2)$.

c. Determine the distribution of X.

(a) Since $\sum_{i=0}^{2} f(x_i) = 1$, we have $K + 2K + 3K = 1 \Rightarrow K = \frac{1}{6}$.

(b) $P(X<2) = P(X=0) + P(X=1) = K + 2K = 3K = \frac{1}{2}$.

$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) = K + 2K + 3K$

$= 6K = 1$.

$P(0<X<2) = P(X=1) = 2 \cdot \frac{1}{6} = \frac{1}{3}$.

distribution

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{6} & \text{if } 0 \leq x < 1. \\ K + 2K = \frac{1}{6} + 2 \cdot \frac{1}{6} = \frac{1}{2} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

$$= \begin{cases} 0 & \text{if } x < 0 \\ 1/6 & \text{if } 0 \leq x < 1 \\ 1/2 & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2. \end{cases}$$

Prob: Let $P(x)$ be the probability function of a discrete random variable 'X' which assumes the values x_1, x_2, x_3, x_4 Such that $2P(x_1) = 3P(x_2) = P(x_3) = 5P(x_4)$.

Find the probability distribution and Cumulative distribution function

Sol: $2P(x_1) = 3P(x_2) = P(x_3) = 5P(x_4) = A$.

$P(x_1) = A/2$, $P(x_2) = A/3$, $P(x_3) = A$ $P(x_4) = \frac{A}{5}$

Also we know that, total probability is 1.

$$\frac{A}{2} + \frac{A}{3} + A + \frac{A}{5} = 1$$ i.e, $P(x_1) + P(x_2) + P(x_3) + P(x_4) = 1$.

$$A(1/2 + 1/3 + 1 + 1/5) = 1.$$

$$A\left(\frac{15+10+30+6}{30}\right) = 1 \Rightarrow A = \frac{30}{61}.$$

∴ The probability distribution of X

$x = x_i$	x_1	x_2	x_3	x_4
$P(X=x_i) =$	$15/61$	$\frac{10}{61}$	$\frac{30}{61}$	$6/61$.
$F(X=x_i)$	$15/61$	$\frac{25}{61}$	$\frac{55}{61}$	$\frac{61}{61} = 1$.

Continuous Probability Distribution.

When a random variable X takes every value in an interval, it gives rise to continuous distribution of X. The distributions defined by the varieles like temp, heights and weights are continuous distribution.

Probability Density Function: For continuous variable, the probability distribution is called Probability Density function because it is defined for every point in the range and not only for the certain values.

Consider the small interval $\left[x-\frac{dx}{2}, x+\frac{dx}{2}\right]$ of length dx around the point x. Let $f(x)$ be any continuous function of x so that $f(x)dx$ represents the probability that the variable X falls in infinitesimal interval $\left[x-\frac{dx}{2}, x+\frac{dx}{2}\right]$. Symbolically it can be expressed as

$P\left[x-\frac{dx}{2} \leq X \leq x+\frac{dx}{2}\right] = f(x)dx$. Then $f(x)$ is called probability density function or simply density function of the variate X and the continuous curve $y=f(x)$ is known as the probability density curve or simply Probability Curve.

In other words: the probability density function of a continuous random variable 'X' denoted by $f(X=x)$ or $f(x)$ is simple way and it satisfies following two properties

* $0 \leq f(x) \leq 1$ for $\forall x \in R$.
* $\int_{-\infty}^{\infty} f(x)dx = 1$.

As the probability for a variate value to lie in the interval dx is $f(x)\,dx$, so the probability for a variate value to fall in the interval (a, b) is $\int_a^b f(x)\,dx$ which represents the area between the curve $y = f(x)$, the X-axis and the ordinates $x = a$ and $x = b$. Since the total probability is unity, we have $\int_a^b f(x)\,dx = 1$, where $[a, b]$ is the range of the variate X.

The range of the variable may be finite or infinite. But even when the range is finite, it is convenient to consider it as infinite by supporting the density function to zero out side the given interval.

Properties of the Probability Density function $f(x)$

1. $f(x) \geq 0 \quad \forall x \in R$ ii $\int_{-\infty}^{\infty} f(x)\,dx = 1$

The probability $P(E)$ is given by

$P(E) = \int_E f(x)\,dx$ is well defined for any event E.

In the case of continuous random variable, we associate the probabilities with intervals. In this case the probability of the variable at a particular point is always zero.

That is, inclusion or non-inclusion of end points, does not change the probability, which is not the case in the discrete-distribution.

Cummulative Distribution Function of A Continuous Random Variable :

The Cummulative distribution function or simply distribution function of a Continuous random variable X is denoted by $F(x)$ and defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^{\infty} f(x).$$

Thus $F(x)$ gives the probability that the value of the variable X will be $\leq x$.

Properties

1. $0 \leq F(x) \leq 1 \quad -\infty < x < \infty$
2. $F'(x) = f(x) \geq 0$, So $F(x)$ is non-decreasing fn.
3. $F(-\infty) = 0$, 4) $F(\infty) = 1$.
5. $F(x)$ is continuous of $F(x)$ are countable.
6. $P(a \leq X \leq b) = \int_a^b f(x)dx = F(b) - F(a)$.

Since $F'(x) = f(x)$ we have $\frac{d}{dx}(F(x)) = f(x) \Rightarrow$

$dF = f(x)dx$ This is known as probability differential of X.

Measures of Central Tendancy for Continuous Probability Distribution.

If a variable is continuous then it takes all possible values in its range like height of an individual, life of a car battery, etc.

The expectation (mean) of the variable is defined as

$$Ex = \int x f(x)\, dx$$ where $f(x)$ is the probability function of the variable x.

Let $f(x)$ be the probability density function of a continuous random variable X. then

i) Mean: Mean of a distribution is given by

$$\mu = E(x) = \int_{-\infty}^{\infty} x f(x) dx.$$

if X is defined from a to b then

$$\boxed{\mu = E(x) = \int_{a}^{b} x f(x) dx}$$

In general, mean or expectation of any function $\phi(x)$ is given by

$$\boxed{E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x) dx.}$$

$f(x)$ be the probability density function of a random variable X.

(ii) Median: Median is the point which divides the entire distribution into two equal parts. In case of continuous distribution, median is the point which divides the total area into two equal parts. Thus if X is defined from a to b and M is the median, then

$$\int_a^M f(x)\,dx = \int_M^b f(x)\,dx = \frac{1}{2}$$

Solving for M, We get the median.

(iii) Mode: Mode is the value of x for which $f(x)$ is maximum. Mode is thus is given by $f'(x) = 0$ and $f''(x) < 0$ for $a < x < b$.

(iv) Variance: Variance is a distribution is given by

$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx \text{ or } \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

Suppose that the variate X is defined from a to b. then $\sigma^2 = \int_a^b (x-\mu)^2 f(x)\,dx$ or $\int_a^b x^2 f(x)\,dx - \mu^2$

(v) Mean Deviation: Mean deviation about the mean is given by $\int_{-\infty}^{\infty} |x-\mu| f(x)\,dx$.

Prob: If a random variable has the probability density function $f(x)$ as

$$f(x) = \begin{cases} 2e^{-2x}, & \text{for } x > 0 \\ 0 & \text{for } x < 0 \end{cases}$$

find the probabilities that it will take on a value

(i) between 1 and 3 (ii) greater than 0.5.

(i) The probability that a variate takes a value between 1 & 3 is given by

$$P(1 \le x \le 3) = \int_1^3 f(x)\,dx = \int_1^3 2e^{-2x}\,dx = 2\left[\frac{e^{-2x}}{-2}\right]_1^3$$

$$= 1 - (e^{-6} - e^{-1}) = e^{-2} - e^{-6}.$$

(ii). The probability that a variable takes a value greater than 0.5 is

$$P(x \ge 0.5) = \int_{0.5}^{\infty} f(x)\,dx = \int_{0.5}^{\infty} 2e^{-2x}\,dx.$$

$$= 2\left(\frac{e^{-2x}}{-2}\right)_{0.5}^{\infty} = (e^{-\infty} - e^{-1}) = \frac{1}{e} = .$$

Prob: If the probability density of a random variable is given by

$$f(x) = \begin{cases} K(1-x^2), & \text{for } 0<x<1 \\ 0 & \text{otherwise.} \end{cases}$$

find the value of K and the probabilites that a random variable having this probability density will take on a value i) between 0.1 and 0.2
ii) greater than 0.5

Sol: Given $f(x) = \begin{cases} K(1-x^2), & \text{for } 0<x<1 \\ 0 & \text{otherwise.} \end{cases}$

We know that $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

i.e., $\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx = 1.$

$$0 + \int_0^1 K(1-x^2)\,dx + 0 = 1.$$

$$K\int_0^1 (1-x^2)\,dx = K\left(x - \frac{x^3}{3}\right)_0^1 = 1.$$

$$= K\left(1 - \tfrac{1}{3}\right) = 1 \Rightarrow \tfrac{2}{3}K = 1.$$

$$\therefore K = 3/2$$

(ii) The probability that the variate will take on a value between 0.1 and 0.2 is.

$$P(0.1<x<0.2) = \int_{0.1}^{0.2} f(x)\,dx = \int_{0.1}^{0.2} K(1-x^2)\,dx. \quad \because K = 3/2$$

$$= \frac{3}{2}\int_{0.1}^{0.2} (1-x^2)\,dx = \frac{3}{2}\left(x - x^3/3\right)_{0.1}^{0.2}$$

$$= \frac{3}{2}\left(0.2 - \frac{0.008}{3} - 0.1 + \frac{0.001}{3}\right)$$

$$= \frac{3}{2}\left(0.1 - \frac{0.007}{3}\right) = 0.2965.$$

(ii) The probability that the variate will take on a value greater than 0.5 is.

$$P(X > 0.5) = \int_{0.5}^{\infty} f(x)\,dx = \int_{0.5}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx.$$

$$= \frac{3}{2}\int_{0.5}^{1} (1-x^2)\,dx + 0 = \frac{3}{2}\left(x - x^3/3\right)_{0.5}^{1}$$

$$= \frac{3}{2}\left[(1 - 1/3) - \left(0.5 - \frac{0.125}{3}\right)\right]$$

$$= \frac{3}{2}\left[(1-1/3) - (0.5 - 0.0417)\right] = 0.3125.$$

$$= \frac{3}{2}(2/3 - 0.4583) = 0.3125.$$

=

Prob: Probability density function of a rand variable x is

$$f(x) = \begin{cases} \frac{1}{2}\sin x & 0 \le x \le \pi \\ 0 & \text{otherwise.} \end{cases}$$

Find the mean, mode and median of the distribution and also find the probability between 0 and $\pi/2$.

Sol. (i) Mean of the distribution $= \int_{-\infty}^{\infty} x f(x)\,dx$.

$$= \int_{-\infty}^{0} x f(x) + \int_{0}^{\pi} x \cdot f(x)\,dx + \int_{\pi}^{\infty} x f(x)\,dx .$$

$$= \int_{-\infty}^{\infty} x \cdot 0 + \int_{0}^{\pi} x \cdot \frac{1}{2}\sin x\,dx + \int_{\pi}^{\infty} x(0)\,dx .$$

$$= \frac{1}{2}\int_{0}^{\pi} x \sin x\,dx = \frac{1}{2}\left(-x\cos x + \sin x\right)_0^{\pi}.$$

$$= \frac{\pi}{2}$$

(ii) Mode is the value of x for which f(x) is maximum

Now $f'(x) = \frac{1}{2}\cos x$.

for f(x) to be maximum $f'(x) = 0$.

i.e., $\cos x = 0$

$\therefore x = \pi/2$.

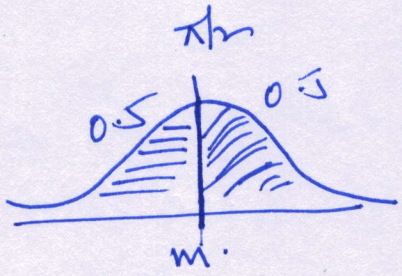
$f''(x) = -\frac{1}{2}\sin x$. At $x = \pi/2$, $f''(x) \cdot 2 = -\frac{1}{2} < 0$.

Hence f(x) is maximum at $x = \pi/2$.

$\therefore$ Mode of the distribution is given by $x = \frac{\pi}{2}$.

If M is the median of the distribution, then.

$$\int_a^M f(x)dx + \int_M^b f(x)dx = \frac{1}{2}.$$



i.e., $\int_0^M \frac{1}{2}\sin x\,dx = \int_M^{\pi} \frac{1}{2}\sin x\,dx = \frac{1}{2}.$

Solving $\int_0^M \frac{1}{2}\sin x\,dx = -\frac{1}{2}(\cos x)_0^M = \frac{1}{2}.$

$$-\frac{1}{2}(\cos M - \cos 0) = \frac{1}{2}$$

$$1 - \cos M = 1 \Rightarrow \cos M = 0 = \cos\frac{\pi}{2}$$

$$\therefore M = \frac{\pi}{2}$$

$\therefore$ Median of the distribution $= \frac{\pi}{2}$.

Thus Mean = Median = Mode $= \frac{\pi}{2}$.

iv $P(0 < x < \pi/2) = \int_0^{\pi/2} f(x)dx = \int_0^{\pi/2} \frac{1}{2}\sin x\,dx.$

$$= -\frac{1}{2}(\cos x)_0^{\pi/2} = -\frac{1}{2}(0-1) = \frac{1}{2}.$$

$=$.

Prob: A continuous random variable has the probability density function

$$f(x) = \begin{cases} Kxe^{-\lambda x} & \text{for } x \geq 0, \lambda > 0. \\ 0 & \text{otherwise} \end{cases}$$

Determine K, mean, variance.

Sol. The total probability is unity, we have

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

$$\text{i.e., } \int_{-\infty}^{0} 0 \cdot dx + \int_{0}^{\infty} Kx e^{-\lambda x}\,dx = 1$$

$$K \int_{0}^{\infty} x e^{-\lambda x}\,dx = 1.$$

$$\text{i.e., } K\left[x \cdot \frac{e^{-\lambda x}}{-\lambda} - 1 \cdot \frac{e^{-\lambda x}}{\lambda^2}\right]_0^{\infty} = 1.$$

$$K\left[(0 - 0 - (0 - 0 \cdot \frac{1}{\lambda^2})\right] = 1. \Rightarrow \frac{K}{\lambda^2} = 1 \quad K = \lambda^2.$$

Now $f(x)$ becomes.

$$= \begin{cases} \lambda^2 x e^{-\lambda x} & \text{for } x \geq 0, \lambda > 0. \\ 0 & \text{otherwise.} \end{cases}$$

Mean of the distribution $\mu = \int_{-\infty}^{\infty} x f(x)\,dx.$

$$\text{i.e., } \mu = \int_{-\infty}^{0} 0 \cdot dx + \int_{0}^{\infty} x f(x)\,dx +$$

$$= \int_{0}^{\infty} x \cdot \lambda^2 x e^{-\lambda x}\,dx. = \int_{0}^{\infty} x^2 \lambda^2 e^{-\lambda x}\,dx.$$

Using Bernoulli's Rule.

$$= \lambda^2\left[(0-0+0)-(0-0-2/\lambda^3)\right] = 2/\lambda .$$

Variance of the distribution.

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2 .$$

i.e., $\sigma^2 = \int_0^{\infty} x^2 f(x)\,dx - (2/\lambda)^2 = \lambda^2 \int_0^{\infty} x^3 e^{-\lambda x}\,dx - \frac{4}{\lambda^2}$

$$= \lambda^2\left[x^3\left(\frac{e^{-\lambda x}}{-\lambda}\right) - 3x^2\left(\frac{e^{-\lambda x}}{\lambda^2}\right) + 6x\left(\frac{e^{-\lambda x}}{-\lambda^3}\right) - 6\left(\frac{e^{-\lambda x}}{\lambda^4}\right)\right]_0^{\infty} - (4/\lambda^2)$$

$$= \lambda^2\left[(0 - 0 + 0 \cdot 0 - 0) - \left(0 - 0 + 0 - \frac{6}{\lambda^4}\right)\right] - \frac{4}{\lambda^2}$$

$$= \frac{6}{\lambda^2} - \frac{4}{\lambda^2} = \underline{\frac{2}{\lambda^2}}.$$

Prob: Is the function defined by

$$f(x) = \begin{cases} 0 & x < 2 \\ \frac{1}{18}(2x+3), & 2 \le x \le 4 \\ 0 & x > 4 \end{cases}$$

a probability density function. Find the probability that variate having $f(x)$ as density function will fall in the interval $2 \le x \le 3$.

For all points x in $-\infty \le x \le \infty$, $f(x) \ge 0$ and

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{2} f(x) + \int_{2}^{4} f(x)\,dx + \int_{4}^{\infty} f(x)\,dx .$$

$$= \int_{-\infty}^{2} 0 \cdot dx + \int_{2}^{4} \frac{1}{18}(2x+3)\,dx + \int_{4}^{\infty} 0 \cdot dx .$$

$$= \frac{1}{18}\int_2^4 (2x+3)\,dx = \frac{1}{18}\left[\frac{(2x+3)^2}{4}\right]_2^4$$

$$= \frac{1}{72}(121-49) = 1.$$

Hence $f(x)$ is a probability density function.

The probability that the density will fall in the interval $2 \le x \le 3$ is

$$P(2 \le x \le 3) = \int_2^3 f(x)\,dx = \frac{1}{18}\int_2^3 (2x+3)\,dx.$$

$$= \frac{1}{18}\cdot\left(x^2+3x\right)_2^3 = \frac{1}{18}(18-10) = \frac{8}{18} = \frac{4}{9}$$

Prob: A random variable X gives measurements x between 0 and 1 with a probability function $f(x) = \begin{cases} 12x^3-21x^2+10x & \text{for } 0 \le x \le 1 \\ 0 & \text{elsewhere.} \end{cases}$

1. find $P(X \le 1/2)$ and $P(X > 1/2)$.

ii find the number k such that $P(X \le K) = 1/2$.

(i) $P(X \le 1/2) = \int_0^{1/2} f(x)\,dx = \int_0^{1/2} (12x^3-21x^2+10x)\,dx$

$$= \left(12\cdot\frac{x^4}{4} - 21\cdot\frac{x^3}{3} + 10\cdot\frac{x^2}{2}\right)_0^{1/2} = \left(3x^4-7x^3+5x^2\right)_0^{1/2}$$

$$= 9/16.$$

$$P(X > 1/2) = 1 - P(x \le 1/2) = 1 - 9/16 = \frac{7}{16}.$$

(ii) Given $P(X \le k) = 1/2$ i.e., $\int_0^k f(x)\,dx = 1/2$

i.e., $\int_0^k (12x^3-21x^2+10x)\,dx = 1/2 \Rightarrow \left(3x^4-7x^3+5x^2\right)_0^k = 1/2$

$$3k^4-7k^3+5k^2 = 1/2. \Rightarrow 6k^4-14k^3+10k^2-1 = 0$$

$$k = 0.452.$$

Prob: A Continuous random variable X has distribution functio

$$F(x) = \begin{cases} 0 & \text{if } x \le 1 \\ k(x-1)^4 & \text{if } 1 < x \le 3 \\ 1 & \text{if } x > 3. \end{cases}$$

Determine $f(x)$, k and Mean.

We knows that probability density function

$$f(x) = \frac{d}{dx}[F(x)] = f(x) \begin{cases} 0 & \text{if } x \le 1. \\ 4k(x-1)^3 & \text{if } 1 < x \le 3 \\ 0 & \text{if } x > 3. \end{cases}$$

Since the total probability is unity, we have

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

$$\int_1^3 4k(x-1)^3\,dx = 1 = \left[4k \frac{(x-1)^4}{4} \right]_1^3 = 1.$$

$$4k\left(\frac{16}{4}\right) = 1 \Rightarrow k = \frac{1}{16}.$$

Hence $f(x) = \begin{cases} 0, & \text{if } x \le 1. \\ \frac{1}{4}(x-1)^3, & \text{if } 1 < x \le 3. \\ 0 & \text{if } x > 3. \end{cases}$

(ii) Mean of X is $\mu = \int_{-\infty}^{\infty} x f(x)\,dx.$

$$= \int_{-\infty}^{1} x f(x)\,dx + \int_1^3 x f(x)\,dx + \int_3^{\infty} x \cdot f(x)\,dx.$$

$$= 0 + \int_1^3 x \cdot \frac{1}{4}(x-1)^3\,dx + 0$$

$$= \frac{1}{4}\int_1^3 x(x-1)^3\,dx.$$

Put $x - 1 = t$.

$$= \frac{1}{4} \int_0^2 (t+1) t^3 \, dt.$$

$$= \frac{1}{4} \int_0^2 t^4 + t^3 \, dt = \frac{1}{4} \left(\frac{t^5}{5} + \frac{t^4}{4} \right)_0^2.$$

$$= \frac{1}{4} \left(\frac{2^5}{5} + \frac{2^4}{4} \right) = \frac{2^4}{4} \left(\frac{2}{5} + \frac{1}{4} \right)$$

$$= 4 \cdot \frac{13}{20} = \frac{13}{5} = 2.6$$

1.(a) The probability density function of a variate X is as follows :

$X = x$	0	1	2	3	4	5	6
$P(x)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$

(*i*) Find $P(X < 4)$, $P(X \geq 5)$, $P(3 < X \leq 6)$

(*ii*) What will be the minimum value of K so that $P(X \leq 2) > 0.3$.

[JNTU (K) Nov. 2009, Mar. 2014 (Set No. 4)]

(*b*) A random variable X has the following probability function :

X	0	1	3	4	5	6	7
$P(X)$	0	K	$2K$	$2K$	$3K$	K^2	$7K^2 + K$

(*i*) Find the value of K (*ii*) Evaluate $P(X < 6), P(X \geq 6)$

(*iii*) Evaluate $P(0 < X < 5)$ **[JNTU (K) Mar. 2013, II Sem June 2015 (Set No. 3)]**

(*c*) A discrete random variable X has the following probability distribution

Value of X	0	1	2	3	4	5	6	7	8
$P(X = x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$	$15k$	$17k$

(*i*) Find the value of 'k' (*ii*) Find $P(X \leq 3), P(0 < X < 3), P(X \geq 3)$

[JNTU (K) Mar. 2014 (Set No.1)]

(*d*) A discrete random variable X has the following probability distribution

Value of X	1	2	3	4	5	6	7	8
$P(X = x)$	$2k$	$4k$	$6k$	$8k$	$10k$	$12k$	$14k$	$4k$

(*i*) Find the value of 'k' (*ii*) Find $P(X < 3)$ and $P(X \geq 5)$

(*iii*) Find the distribution function of X. **[JNTU (K) Mar. 2014 (Set No. 3)]**

2. A random varibale X has the following probability function :

$X = x$	-2	-1	0	1	2	3
$P(X)$	0.1	K	0.2	$2K$	0.3	K

Find (*i*) K (*ii*) Mean (*iii*) Variance (*iv*) $P(x \geq 2)$ (*v*) $P(x < 2)$

(*vi*) $P(-1 < x < 3)$ **[JNTU 2007S, (A) May 2012, (K) II Sem. June 2015 (Set No. 2)]**

3. A random variable X has the following probability function :

$X = x$	4	5	6	8
$P(X)$	0.1	0.3	0.4	0.2

Determine (*i*) Expectation (*ii*) Variance (*iii*) Standard deviation

4. Find mean and variance for the following discrete

5. (*a*) A random variable X has the following probability function :

$X = x$	1	2	3	4	5	6
$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$

Determine (*i*) K (*ii*) Expectation (*iii*) Variance

(*b*) A random variable X has the following probability function :

X	0	1	2	3	4	5	6	7	8
$P(X)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

(*i*) Determine the value of a,

(*ii*) Evaluate $P(X < 3), P(X \geq 3), P(2 \leq X < 5)$ **[JNTU (K) Nov. 2012 (Set No. 2)]**

6. Find the expected value of x and standard deviation for the following discrete distribution :

X	8	12	16	20	24
$P(X)$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{12}$

[JNTU Jan. 2007]

7. A random variable X has the following probability function :

X	0	1	2	3	4	5	6	7	8
$P(X)$	$\frac{K}{45}$	$\frac{K}{15}$	$\frac{K}{9}$	$\frac{K}{5}$	$\frac{2K}{45}$	$\frac{6K}{45}$	$\frac{7K}{45}$	$\frac{8K}{45}$	$\frac{4K}{45}$

Determine (*i*) K (*ii*) Mean (*iii*) Variance (*iv*) Standard deviation.

[JNTU (K) Nov. 2011 (Set No. 3)]

8. For the following probability distribution

X	-3	-2	-1	0	1	2	3
$P(X)$	0.001	0.01	0.1	?	0.1	0.01	0.001

Find (*i*) the missing probability (*ii*) mean

(*iii*) variance (*iv*) $E(X^2 + 2X + 3)$

9. A fair coin is tossed until a head or five tails occurs.

Find (*i*) the discrete probability distribution (*ii*) mean of the distribution.

10. A box contains 8 items of which 2 are defective. A man draws 3 items from the box. Find the expected number of defective items he has drawn.

11. A die is thrown at random. What is the expectation of the number on it ?

12. A box contains 4 white and 6 black balls. A man draws 2 balls and is given Rs. 140 for every white ball and Rs. 70 for every black ball. What is his expectation ?

13. Five defective bolts are accidentally mixed with 20 good ones. Find the probability distribution of the number of defective bolts, if four bolts are drawn at random from this lot.

14. Two dice are rolled at random. Find the probability distribution of the sum of the numbers on them. Also find the mean of the distribution.

15. If 3 cars are selected from a lot of 6 cars containing 2 defective cars, find the probability distribution of the number of defective cars. **[JNTU 2008S (Set No.1)]**

16. Find the expected value and variance of the following distribution :

X	-10	-20	30	75	80
$p(x)$	$\frac{1}{5}$	$\frac{3}{20}$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{20}$

17. The probability distribution of a random variable X is given below :

X	-2	-1	0	1	2
$p(x)$	0.2	0.1	0.3	0.3	0.1

Find (*i*) $E(X)$ (*ii*) Var (X) (*iii*) $E(2X-3)$ (*iv*) Var $(2X-3)$

18. Let X denotes the minimum of the two numbers that appear when a pair of fair dice is thrown once. Determine the (*i*) Discrete probability distribution (*ii*) Expectation (*iii*) Variance. **[JNTU (K) May 2013 (Set No.2)]**

ANSWERS

1. (*a*) (*i*) $\frac{33}{49}$ (*ii*) $\frac{1}{30}$ **2.** (*i*) 0.1 (*ii*) 0.8 (*iii*) 2.16
3. (*i*) 5.9 (*ii*) 1.49 (*iii*) 1.22 **4.** 0.8, $\frac{143}{50}$
5. (*a*) (*i*) $\frac{1}{36}$ (*ii*) 4.46 (*iii*) 2.08 **6.** $16, 2\sqrt{5}$
7. (*i*) 1 (*ii*) 4.622 (*iii*) 4.9971 (*iv*) 2.24
8. (*i*) 0.778 (*ii*) 0.2 (*iii*) 0.258 (*iv*) 3.698
9.

x	1	2	3	4	5
$f(x)$	1/2	1/4	1/8	1/16	1/16

; 1.9

10. $\frac{3}{4}$ **11.** $\frac{7}{2}$ **12.** 196

13.

x	0	1	2	3	4
$p(x)$	$\frac{969}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

14.

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$p_i = p(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

15.

x	0	1	2
$p(x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{2}{5}$

16. 21.5, 950.24 **17.** (*i*) 0 (*ii*) 1.6 (*iii*) -3 (*iv*) 6.4

EXERCISE 2(B)

1. The probability density function $f(x)$ of a continuous random variable is given by

$$f(x) = \begin{cases} k\,x^3, \text{ for } 0 < x < 1 \\ 0, \text{ elsewhere} \end{cases}$$

Find the value of k and the probability that the random variable takes on a value

(*i*) between $\frac{1}{4}$ and $\frac{3}{4}$ (*ii*) greater than $\frac{2}{3}$.

2. (*a*) If the probability density of a random variable is given by

$$f(x) = \begin{cases} x, \text{ for } 0 < x < 1 \\ 2 - x, \text{ for } 1 \le x < 2 \\ 0, \text{ elsewhere} \end{cases}$$

Find the probabilities that a random vairable having this probability density will take on a value (*i*) between 0.2 and 0.8 (*ii*) between 0.6 and 1.2.

[JNTU (K) Mar. 2014 (Set No.2)]

(*b*) If the probability density of a random variable is given by $f(x) = \begin{cases} kx^2, 0 < x < \\ 0, \text{elsewhere} \end{cases}$

Find the value of *k* and the probabilities that a random variable having this probability density will take on a value (*i*) between 0.25 and 0.75, (*ii*) greater than 2/3.

[JNTU (K) Mar. 2014 (Set No. 1)]

(*c*) If the probability density of a random variable X is given by

$$f(x) = \begin{cases} k(1-x^2), 0 < x < 1 \\ 0, \text{otherwise} \end{cases}$$

Find (*i*) *k* and (*ii*) The cumulative distribution function of X.

3. Show that the function $f(x) = \frac{1}{\pi(1+x^2)}$, $-\infty \le x \le \infty$ is a probability density function.

[JNTU (K) Nov. 2011 (Set No. 4)]

4. Find the constant *k* so that the function $f(x)$ defined by

$$(i)\ f(x) = \begin{cases} \frac{1}{k}, \text{if } a \le x \le b \\ 0, \text{elsewhere} \end{cases} \qquad (ii)\ f(x) = \begin{cases} k\,e^{-3x}, x > 0 \\ 0, \text{elsewhere} \end{cases}$$

is a density function.

5. (*a*) A random variable X has the following probability function :

$X = x$	1	2	3	4	5	6
$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$

Determine (*i*) K (*ii*) Expectation (*iii*) Variance

(*b*) A random variable X has the following probability function :

X	0	1	2	3	4	5	6	7	8
$P(X)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

(*i*) Determine the value of a,

(*ii*) Evaluate $P(X < 3), P(X \geq 3), P(2 \leq X < 5)$ **[JNTU (K) Nov. 2012 (Set No. 2)]**

6. Find the expected value of x and standard deviation for the following discrete distribution :

X	8	12	16	20	24
$P(X)$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{12}$

[JNTU Jan. 2007]

7. A random variable X has the following probability function :

X	0	1	2	3	4	5	6	7	8
$P(X)$	$\frac{K}{45}$	$\frac{K}{15}$	$\frac{K}{9}$	$\frac{K}{5}$	$\frac{2K}{45}$	$\frac{6K}{45}$	$\frac{7K}{45}$	$\frac{8K}{45}$	$\frac{4K}{45}$

Determine (*i*) K (*ii*) Mean (*iii*) Variance (*iv*) Standard deviation.

[JNTU (K) Nov. 2011 (Set No. 3)]

8. For the following probability distribution

X	– 3	– 2	– 1	0	1	2	3
$P(X)$	0.001	0.01	0.1	?	0.1	0.01	0.001

Find (*i*) the missing probability (*ii*) mean

(*iii*) variance (*iv*) $E(X^2 + 2X + 3)$

9. A fair coin is tossed until a head or five tails occurs.

Find (*i*) the discrete probability distribution (*ii*) mean of the distribution.

10. A box contains 8 items of which 2 are defective. A man draws 3 items from the box. Find the expected number of defective items he has drawn.

11. A die is thrown at random. What is the expectation of the number on it ?

12. A box contains 4 white and 6 black balls. A man draws 2 balls and is given Rs. 140 for every white ball and Rs. 70 for every black ball. What is his expectation ?

13. Five defective bolts are accidentally mixed with 20 good ones. Find the probability distribution of the number of defective bolts, if four bolts are drawn at random from this lot.

14. Two dice are rolled at random. Find the probability distribution of the sum of the [illegible] Also find the mean of the distribution.

15. If 3 cars are selected from a lot of 6 cars [illegible] probability distribution of the number of defective cars. [JNTU 2008S (Set No.]

16. Find the expected value and variance of the following distribution :

X	−10	−20	30	75	80
$p(x)$	$\frac{1}{5}$	$\frac{3}{20}$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{20}$

17. The probability distribution of a random variable X is given below :

X	−2	−1	0	1	2
$p(x)$	0.2	0.1	0.3	0.3	0.1

Find (*i*) $E(X)$ (*ii*) Var (X) (*iii*) $E(2X-3)$ (*iv*) Var $(2X-3)$

18. Let X denotes the minimum of the two numbers that appear when a pair of fair dice is thrown once. Determine the (*i*) Discrete probability distribution (*ii*) Expectation (*iii*) Variance. [JNTU (K) May 2013 (Set No.2]

ANSWERS

1. (*a*) (*i*) $\frac{33}{49}$ (*ii*) $\frac{1}{30}$ **2.** (*i*) 0.1 (*ii*) 0.8 (*iii*) 2.16
3. (*i*) 5.9 (*ii*) 1.49 (*iii*) 1.22 **4.** 0.8, $\frac{143}{50}$
5. (*a*) (*i*) $\frac{1}{36}$ (*ii*) 4.46 (*iii*) 2.08 **6.** $16, 2\sqrt{5}$
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9.

x	1	2	3	4	5
$f(x)$	1/2	1/4	1/8	1/16	1/16

; 1.9

10. $\frac{3}{4}$ **11.** $\frac{7}{2}$ **12.** 196

13.

x	0	1	2	3	4
$p(x)$	$\frac{969}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

14.

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$p_i = p(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

15.

x	0	1	2
$p(x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{2}{5}$

16. 21.5, 950.24 **17.** (*i*) 0 (*ii*) 1.6 (*iii*) −3 (*iv*) 6.4

So far we have learnt frequency distributions which are based on the actual observations of experperiment.

Now we shall discuss the theoretical distributionis which variates are distributed as per some definite laws which can be expressed Mathematically. That is, theoretical distributions (or Probability distributions) are Mathematical models. So, it is possible to deduce Mathematically what the frequency distribution of certain population should be. Such distributions are called Theoretical distributions or Frequency distributions. Hence a theoritical distribution is the frequency distribution of a certain event in which frequencies are obtained by Mathematical Computation.

There are two types of theoritical distributions (or probability distributions).

1. Discrete Theoretical Distributions.
   a. Binomial Distribution
   b. Poisson Distributions
   c. Rectangular distribution.
   d. Geo. Distribution.
2. Continuous Theoretical Distribution
   a- Normal Distribution.
   b. Chi-Square distribution

Bernoulli's Distribution :

A random variable X which takes two values 0 and 1 with probability q and p respectively i.e., $P(X=0)=q$ $P(X=1)=P$, $q=1-P$ is called Bernoulli's discrete random variable and is said to have Bernoulis distribution.

The probability function of Bernoulli's distribution can be written as.

$$P(x) = p^x q^{1-x} = p^x (1-p)^{1-x}, \quad x = 0, 1 \ldots .$$

Mean of Bernoulis discrete random variable X is

i. $\mu = E(X) = \sum x_i P(x_i) = 0 \times q + 1 \times p = p.$

(ii) $V(x) = E(X^2) - [E(X)]^2 = \sum x_i^2 . P(x_i) - \mu^2$

$= (0^2 \times q + 1^2 \times p) - p^2 = p - p^2 = p(1-p)$

$= pq.$

Hence standard deviation is $\sigma = \sqrt{pq}$..

Bernoulis Theorem:

If the probability of the occurence of an event (success) is a single trial is P. then the probability that it will occur exactly r times out of n independt Bernoulis trials is

$${}^nC_r p^r q^{n-r} \text{ where } p+q=1.$$

Bionomial Distributions:

The theory of bionomial distribution was developed by Bernouli in the year 1700 and it is discrete Probability distribution.

Let n represents the number of trials and x represents the no. of successes then the following three conditions (assumptions should be satisfied.

1. There are only two possible out comes in each trial. Arbitrary called Successes and failure (Here a Successes is not neccessarily a desirable out come).
2. The probability of Successes is the same in each trial (Consequently, the prob. of the failure is also Same in each trial.
3. The n trials are independent. These are Bernouli trials.

Def: A random variable X has a Bionomial distribution if it assumes only non-nagative values and its Probability density function is given by

$$P(X=r) = P(r) = \begin{cases} {}^{n}C_r\, p^r\, q^{n-r} & ; r=0,1,2\ldots n \\ 0 & \text{otherwise.} \end{cases}$$

$q = 1-p$

An alternative notation for a bionomial distribution

$$P(X=x) = P(x) = \begin{cases} {}^{n}C_r\, p^r\, q^{n-x} & x=0,1,\ldots n \\ 0 & \text{otherwise.} \end{cases}$$

The bionomial distribution function is given by

$$F_X(x) = P(X \le x) = \sum_{r=0}^{x} {}^{n}C_r\, p^r\, q^{n-r}$$

Examples of Binomial Distribution

1. The no. of defective bolts in a box containing n bolts.
2. The no. of machines lying idle in a factory having n-machines
3. The no. of Post-graduates in a group of n men.
4. The no. of oil wells yielding natural gas in a group of n wells test drilled.

Conditions of Binomial Distribution:

The Binomial distribution holds if

1. Trials are repeted under identical Conditions for a fixed number of times, Say n times.
2. There are only two possible out comes Sucess & failure for each trial.
3. The prob. of sucess in each trial remains constant and does not change from trial to trial.
4. The trials are independent. ie, the probability of an event in any trial is not affected by the results of any other trial.

Obs: In n trials, the total no. of possible ways of obtaining x successes is ^nC_x, All of which numerically disjoint.

Hence, the required probability of getting x successes in n trials, is given by

$$P(X=x) = P(x) = p^x q^{n-x} + p^x q^{n-x}.$$

$$= {}^nC_x \, p^x q^{n-x}.$$

Obs: We know that, the Binomial expansion of $(x+y)^n$

$\therefore (x+y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_n x^0 y^n$.

$X = x$	$P(X=x)$	$p^x q^{n-x}$.
0	${}^nC_0$	$p^0 q^n$
1	${}^nC_1$	$p^1 q^{n-1}$
2	${}^nC_2$	$p^2 q^{n-2}$
3	${}^nC_3$	$p^3 q^{n-3}$
⋮	⋮	
n	${}^nC_n$	$p^n q^0$.

Also $(q+p)^n = {}^nC_0 q^n p^0 + {}^nC_1 q^{n-1} p^1 + \dots {}^nC_n q^0 p^n$.

Since the probabilities of $P(X=x) = {}^nC_x p^x q^{n-x}$. $x = 0, 1, 2 \dots$ are successive terms of the Binomial expansion $(q+p)^n$. Here it is called Binomial Distribution

§ The total probability of Binomial Distribution

We know that

Total probability of B.D $= \sum_{x=0}^{n} P(X=x)$

$= \sum_{n=0}^{n} {}^nC_x p^x q^{n-x}$.

$= {}^nC_0 p^0 q^n + {}^nC_1 p^1 q^{n-1} + {}^nC_2 p^2 q^{n-2} + \dots {}^nC_n p^0 q^n$

$= (q+p)^n = (1)^n = 1$.

$\therefore$ The total prob. of B.D is 1.

Mean of Binomial Distribution.

We know that mean of B.D $= E(X)$.

$$= \sum_{x=0}^{n} x \cdot P(X=x).$$

$$= \sum_{x=0}^{n} x \cdot {}^{n}C_{x}\, p^{x} q^{n-x}.$$

$$= \sum_{x=0}^{n} x \cdot \frac{n!}{(n-x)!\,x!}\, p^{x} q^{n-x}.$$

$$= \sum_{x=0}^{n} x \cdot \frac{n(n-1)!}{(n-x)!\,x(x-1)!}\, p \cdot p^{x-1} q^{n-x}$$

$$= np \sum_{x=0}^{n} \frac{(n-1)!\, p^{x-1} \cdot q^{[(n-1)-(x-1)]}}{[(n-1)-(x-1)]!\,(x-1)!}$$

$$= np \sum_{x=1}^{n} {}^{n-1}C_{(x-1)}\, p^{(x-1)} \cdot q^{[(n-1)-(x-1)]}$$

$$= np\left[(n-1) \cdot q^{n-1} + (n-1)\, p\, q^{n-2} + \cdots\right]$$

$$= np\,(q+p)^{n-1}.$$

But $p+q=1$

$$\boxed{\mu = np}$$

~~Fast~~

Variance of B.D.

We know that Variance of B.D $= E(X)^2 - [E(X)]^2$

Now Consider

$$E(X^2) = \sum_{x=0}^{n} x^2 P(X=x)$$

$$= \sum_{x=0}^{n} [x(x-1)+x] P(X=x).$$

$$= \sum_{x=0}^{n} x(x-1) P(X=x) + \sum_{x=0}^{n} x P(X=x).$$

$$= \sum_{x=0}^{n} x(x-1) P(X=x) + np$$

$$= \sum_{x=0}^{n} x(x-1) \frac{n!}{(n-x)!x!} p^x . q^{n-x} + np.$$

$$= \sum_{x=0}^{n} x(x-1) \cdot \frac{n(n-1)(n-2)! \cdot p^2 p^{x-2} q^{(n-2)-(x-2)}}{[(n-2)-(x-2)]! (x-2)!} + np$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \cdot (n-2)! \, p^{x-2} \cdot q^{[(n-2)-(x-2)]} + np.$$

$$= n(n-1)p^2 . (p+q)^{n-2} + np$$

$$= n(n-1)p^2 . (1)^{n-2} + np$$

$$= np + n(n-1)p^2 .$$

$$E(X^2) = np + n(n-1)p^2$$

$\therefore$ Variance of B.D $= E(X^2) - [E(x)]^2.$

$$= [np + n^2 p^2 - np^2] - (np)^2$$

$$= np + n^2 p^2 - np^2 - n^2 p^2.$$

$$= np - np^2 = np(1-p)$$

$$= npq.$$

$\therefore np \geq npq$

Variance of B.D = npq . S.D = $\sqrt{\text{Variance}}$
$= \sqrt{npq}$.

Mode of B.D

Mode of the Binomial distribution is the value of x at which $P(x)$ has max. value.

mode = $\begin{cases} \text{integral part of } (n+1)p, \text{ if } (n+1)p \text{ is not an integer} \\ (n+1)p \text{ and } (n+1)p-1, \text{ if } n+1 \text{ is an integer.} \end{cases}$

Recurrence Relation for B.D.

We know that $P(x) = {}^nC_x\, p^x q^{n-x}$. ——— (1)

$\therefore \quad P(x+1) = {}^nC_{x+1}\, p^{x+1} q^{n-(x+1)}$ ——— (2)

$2 \div 1.$ $\quad \dfrac{P(x+1)}{P(x)} = \dfrac{{}^nC_{x+1}}{{}^nC_x} \cdot \dfrac{p^{x+1} \cdot q^{n-x-1}}{p^x q^{n-x}} = \dfrac{n-r}{r+1} \cdot \dfrac{p}{q}$

or $\quad \boxed{P(x+1) = \dfrac{n-r}{r+1} \cdot \dfrac{p}{q} \cdot P(x)}$.

~~Bob~~.

Prob: Ten coins are tossed (thrown) simultaneously, Find the probability of getting at least.

(i) Seven Heads (ii) Six Head

let p = Prob. of getting a head $= \frac{1}{2}$

q = " not getting a Head $= \frac{1}{2}$.

The prob. of getting r heads in a throw of 10 coins is $n = 10$

$$P(X=r) = P(r) = {}^{10}C_r \left(\tfrac{1}{2}\right)^{10-r} \left(\tfrac{1}{2}\right)^{r}$$

$$= {}^{10}C_r \left(\tfrac{1}{2}\right)^{10-r} \left(\tfrac{1}{2}\right).$$

$$= {}^{n}C_r\, p^r \cdot q^{n-r} \qquad r = 0, 1, 2 \ldots 10.$$

(i) Prob. of a getting at least seven heads is given by. $P(X \geq 7)$.

$$= P(X=7) + P(X=8) + P(X=9) + P(X=10)$$

$$= {}^{10}C_7 \left(\tfrac{1}{2}\right)^7 \left(\tfrac{1}{2}\right)^3 + {}^{10}C_8 \left(\tfrac{1}{2}\right)^8 \left(\tfrac{1}{2}\right)^2 +$$

$${}^{10}C_9 \left(\tfrac{1}{2}\right)^9 \left(\tfrac{1}{2}\right) + {}^{10}C_{10} \left(\tfrac{1}{2}\right)^{10}.$$

$$= \frac{1}{2^{10}} \left[{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10} \right].$$

$$= \frac{1}{2^{10}} \left[120 + 45 + 10 + 1 \right] = \frac{176}{1024} = 0.1719$$

(ii) The prob. of getting atleast six heads are given by $P(X \geq 6)$.

$$P(X \geq 6) = P(X=6) + P(X \geq 7)$$

$$= {}^{10}C_6 \left(\tfrac{1}{2}\right)^6 \left(\tfrac{1}{2}\right)^4 + 0.1719$$

$= \frac{210}{1024} + 0.1719 = 0.3769$.

Prob: Out of 800 families with 5 children each, how many would you expect to have a). 3 boys (b). 5 girls c. either 2 or 3 boys d atleast one boys [Assume equal probabilities for boys and girls]

Sol: Let the number of boys in each family $= x$

P = Prob. of each boy $= \frac{1}{2}$ (since equal prob. of boys & girls)

number of children $n = 5$.

The prob. distribution is

$$P(x) = {}^{n}C_x \, p^x q^{n-x} = {}^{5}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$= \frac{1}{2^5} \cdot {}^{5}C_x \text{ per family.}$$

Now (a). $P(3 \text{ boys}) = P(x=3) = P(3) = \frac{1}{2^5} \cdot {}^{5}C_3$.

$= \frac{10}{32} = \frac{5}{16}$ per family

Thus for 800 families the prob. of number of families having 3 boys.

$\frac{5}{16} \times 800 = 250$ families.

(b). $P(5 \text{ girls}) = P(\text{no boys}) = P(x=0) = P(0)$

$= \frac{1}{2^5} \cdot {}^{5}C_0 = \frac{1}{32}$ per family

Thus for 800 families, the probability of the no. of families having 5 girls $= \frac{1}{32} \cdot 800 = 25$ families

(c) P(Either 2 or 3 boys) = P(x=2) + P(x=3).

$= P(2) + P(3)$

$= \frac{1}{25} {}^5C_2 + \frac{1}{25} {}^5C_3$.

$= \frac{1}{25}[10+10] = \frac{20}{32} = 5/8$ per family

∴ Expected no. of families with 2 or 3 boys

$= \frac{5}{8} \times 800 = 500$ families.

(d). P(at least one boy) = P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5).

$= \frac{1}{25}.({}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5)$

$= \frac{1}{25}(5+10+10+5+1) = \frac{31}{32}$

∴ Expected number of families with at least one boy $= \frac{31}{32} \times 800 = \underline{775}$.

Prob: The mean and variance of binomial distribution are 4 and 4/3 respectively find P(X ≥ 1).

Sol. Mean of the binomial distribution = 4

i.e., $np = 4$. — (1)

and variance of binomial distribution $= \frac{4}{3}$

i.e., $npq = 4/3$. — (2)

from 1 & 2 $q = \frac{1}{3}$ $p = 1-q = 1-1/3 = 2/3$.

from 1. $n = \frac{4}{p} = \frac{4}{\frac{2}{3}} = 6$.

∴ $P(X \geq 1) = 1 - P(X < 1)$

$= 1 - P(X=0) = 1 - {}^6C_0 \cdot p^0 \cdot q^{6-0}$

$= 1 - (1/3)^6 = 0.9986$

Prob.

The mean and variance of binomial variable X with two parameters n and p are 16 and 8

Find $P(X \geq 1)$ and $P(X > 2)$.

Sol. The mean of B.D $= 16$ i.e., $np = 16$. — ①

The variance of B.D $= 8$ i.e., $npq = 8$. — ②

from these two. $2 \div 1 \cdot \frac{npq}{np} = q = \frac{8}{16} = \frac{1}{2}$.

$\therefore p = 1 - q = 1/2$. and $n = \frac{16}{p} = \frac{16}{1/2} = 32$

Hence $P(X \geq 1) = 1 - P(X < 1)$

$= 1 - P(X = 0) = 1 - {}^{32}C_0 p^0 \cdot q^{32}$

$= 1 - {}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32} = 1 - \frac{1}{2^{32}}$.

$P(X > 2) = 1 - P(X \leq 2)$.

$= 1 - [P(X=0) + P(X=1) + P(X=2)]$.

$= 1 - \left[{}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32} + {}^{32}C_1 \frac{1}{2} \cdot \left(\frac{1}{2}\right)^{31} + {}^{32}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{30}\right]$.

$= 1 - \left(\frac{1}{2}\right)^{32} \left[{}^{32}C_0 + {}^{32}C_1 + {}^{32}C_2\right]$.

$= 1 - \frac{1}{2^{32}} [1 + 32 + 496] = 0.999$

Prob: 20% of items produced from a factory are defective find the probability that in sample of 5 chosen randomly (i) none is defective (ii) one is defective (iii) $P(1<x<4)$.

Sol: Probability of defective items $= p = 20\% = 0.2$

Probability of non defective items $= q = 1-p = 1-0.2 = 0.8$.

Total no. of items $n=5$.

(i) Prob. of none is defective = prob. of 0 defective items

$= P(0) = {}^5C_0 (0.2)^0 (0.8)^5$.

$= 0.32768$.

(ii) Prob. of one defective item $P(1) = {}^5C_1 (0.2)^1 (0.8)^4$.

$= 5(0.2)(0.4096) = 0.4096$.

(iii) Prob. of $(1<x<4) = P(2) + P(3)$.

$= {}^5C_2 (0.2)^2 (0.8)^3 + {}^5C_3 (0.2)^3 (0.8)^2$

$= (0.2)^2 \cdot (0.8)^2 [10 \cdot (0.8) + 10 \cdot (0.2)]$

$= 0.256$.

Poisson Distribution:

The poisson distribution may be obtained as case of B.D. unless the following conditions

1. The number of trials infinitely large i.e., $n \to \infty$
2. The prob. of success for each trial is infinitely small i.e., $p \to 0$.
3. np is finite quantity. say $np = \lambda$. finite.

Under this conditions the binomial prob. fn.

$P(X=x) = {}^nC_x\, p^x q^{n-x}$ where $n = 0, 1, 2 \ldots$ n tends to

the probability function of poisson Distribution ~~is~~

i.e., $P(X=x) = \frac{\lambda^x}{x!} e^{-\lambda}$ where $x = 0, 1, 2 \ldots \infty$.

is known as probability function of poisson distribution.

Hence limiting case of B.D is P.D.

§ Total probability of P.D.

We know that $\sum_{n=0}^{\infty} P(X=x) = \sum_{n=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda}$.

$$= e^{-\lambda}\left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right]$$

$$= e^{-\lambda} \cdot e^{\lambda} = e^0 = 1. \qquad e^x = 1 + x + \frac{x^2}{2!} + \cdots$$

$\therefore$ Total probability of PD is 1.

Mean of P.D:

We know that mean of P.D. $= E(X)$.

$$= \sum_{x=0}^{\infty} x \cdot P(X=x).$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{\lambda^x}{x!} e^{-\lambda} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{x \cdot \lambda^{x-1} \cdot \lambda}{x(x-1)!}$$

$$= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= \lambda e^{-\lambda} \left[1 + \lambda/1! + \frac{\lambda^2}{2!} \cdots\right]$$

$$= \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda.$$

$E(X) = \lambda = np$. Mean of BD is λ

Variance of P.D: $\boxed{\sigma^2 = \lambda^2 + \lambda}$.

We know that variance of P.D $= E(X^2) - [E(X)]^2$.

Now consider $E(X^2) = \sum_{x=0}^{\infty} x^2 P(X=x)$.

$$= \sum_{x=0}^{\infty} [x(x-1) + x] \cdot \frac{\lambda^x}{x!} e^{-\lambda}.$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x!} \cdot e^{-\lambda} + \sum_{x=0}^{\infty} x \cdot \frac{\lambda^x}{x!} e^{-\lambda}$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x!} e^{-\lambda} + E(X).$$

$$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{x(x-1)\lambda^{x-2}}{x(x-1)(x-2)!} + E(x).$$

$$= e^{-\lambda} \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda.$$ — -ve ! does not exist

$$= e^{-\lambda} \lambda^2 (1 + \lambda/1! + \cdots) + \lambda$$

$$= \lambda^2 e^{-\lambda} \cdot e^{\lambda} + \lambda = \lambda^2 + \lambda.$$

$E(x^2) = \lambda(\lambda+1)$.

Variance of PD $= E(x^2) - [E(x)]^2$

$= \lambda^2 + \lambda - \lambda^2$

$= \lambda$.

mean $np = \lambda$

variance $npq = \lambda$.

$q \Rightarrow 1$ is PD.

for P.D, We have

mean of PD = Variance of PD $= \lambda$.

S.D $\sigma = \sqrt{\lambda}$.

Poisson Distribution is used in many situations such as.

(1) Emisson of radioactive particles.
2. Number of cars passing a certain street at a time t.
3. No. of cars passing a certain street at a time t.
4. No. of faulty blades in a packet of 1000.
5. No. of suicides or deaths due to heart attack in one min
6. No. of accidents that take place on a busy road at time t.

Reccurance Relation of P.D.

We have $P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$, $P(x+1) = \frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}$

$= \frac{e^{-\lambda}\cdot\lambda^x\cdot\lambda}{x!(x+1)}$ $= \frac{\lambda}{x+1}\cdot\frac{e^{-\lambda}\cdot\lambda^x}{x!}$

$$P(x+1) = \frac{\lambda}{x+1}\cdot P(x)$$

Properties:

1. Range of variables is from 0 to ∞
2. Mean and Variance are equal.

Prob: A hospital switch board receives an average of 4 emergency calls in a 10 min interval.

i) What is the probability that there are atmost 2 emergency calls in a 10min. interval.

ii). There are exactly 3 emergency calls in a 10 min interval

Sol. 4 call / 10 minuts

$\therefore$ mean $\lambda = 4$ Calls

$\therefore P(X=x) = P(x) = \frac{e^{-\lambda}\cdot\lambda^x}{x!} = \frac{e^{-4}\cdot 4^x}{x!} = \frac{1}{e^4}\cdot\frac{4^x}{x!}$

(i) P(atmost 2 calls) $= P(X \le 2)$.

$= P(X=0) + P(X=1) + P(X=2)$

$= \frac{1}{e^4} + \frac{1}{e^4}\cdot 4 + \frac{1}{e^4}\cdot\frac{4^2}{2!}$

$= \frac{1}{e^4}(1+4+8) = 13e^{-4} = 0.2381$

(ii) P(Exactly 3 calls) $= P(X=3) = \frac{1}{e^4}\cdot\frac{4^3}{3!} = \frac{32}{3}e^{-4}$

$= 0.1954$

Prob: A bank received on the avg of 6 bad cheques per day. find the prob. that it will receive 4 bad cheques on any given day.

We have $P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}$.

here $\lambda = 6$.

$\therefore P(X=4) = \frac{e^{-6}\cdot 6^4}{4!} = \frac{54}{e^6} = 0.1339$.

Prob: A manufacturer of cotter pins knows that 5% of his product is defective. Pins are sold in boxes of 100. He guarantees that not more than 10 pins will be defective. What is the approximate probability that a box will fail to meet the guaranteed quality.

The probability of cotter pins to be defective

$$p = 5\% = 0.05$$

Total no. of cotter pins $n = 100$.

mean $\lambda = np = 100(0.05) = 5$.

we have $P(X=x) = \frac{e^{-\lambda}\cdot\lambda^x}{x!} = \frac{e^{-5}\cdot 5^x}{x!}$

P (abox will fail to meet the guarantee) $= P(x>10)$

$= 1 - P(x \le 10)$.

$= 1 - P(x \le 10)$.

$= 1 - [P(X=0) + P(X=1) + \cdots P(X=10)]$.

$= 1 - \left[\frac{e^{-5}\cdot 5^0}{0!} + \frac{e^{-5}\cdot 5^1}{1!} + \cdots \frac{e^{-5}\cdot 5^{10}}{10!}\right]$.

$= 1 - e^{-5}\left[1 + 5 + \frac{5^2}{2!} + \cdots \frac{5^{10}}{10!}\right]$

$= 1 - 0.9863 = 0.0137$.

Prob: If a random variable has a poisson distribution such that $P(1) = P(2)$. find i) mean of the distribution (ii) $P(4)$, iii) $P(x \geq 1)$ iv). $P(1 < x < 4)$.

Sol: We know that $P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = P(x)$.

$$P(1) = \frac{e^{-\lambda} \cdot \lambda^1}{1!} \qquad P(2) = \frac{e^{-\lambda} \cdot \lambda^2}{2!}$$

$$P(1) = P(2) \Rightarrow \frac{e^{-\lambda} \cdot \lambda^1}{1!} = \frac{e^{-\lambda} \cdot \lambda^2}{2!} \Rightarrow \lambda = \frac{\lambda^2}{2}.$$

$$2\lambda = \lambda^2$$
$$\lambda^2 - 2\lambda = 0$$
$$\lambda(\lambda - 2) = 0$$
$$\lambda = 0 \text{ or } 2$$

$$\lambda = 2.$$

(ii) $P(4) = P(X=4) = \frac{e^{-\lambda} \cdot \lambda^4}{4!}$

$$= \frac{e^{-2} \cdot 2^4}{4!} = \frac{2}{3} e^{-2} = 0.09022$$

(iii) $P(x \geq 1) = 1 - P(x < 1) = 1 - P(X=0)$

$$= 1 - \frac{e^{-\lambda} \cdot \lambda^0}{0!} = 1 - e^{-2}$$

$$= 1 - \frac{1}{e^2} = 0.8247.$$

(iv) $P(1 < x < 4) = P(x=2) + P(x=3)$.

$$= \frac{e^{-2} \cdot 2^2}{2!} + \frac{e^{-3} \cdot 2^3}{3!}$$

$$= e^{-2}\left(2 + \frac{4}{3}\right) = e^{-2}\left(\frac{10}{3}\right)$$

$$= 0.4511.$$

Prob: If X is a poisson variate such that $P(X=0) = P(X=1)$. find $P(x=0)$ using recurrance formula find the probabilities at $x = 1, 2, 3, 4$ and 5.

Sol: probability distribution is $P(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

Given $P(x=0) = P(x=1)$.

$$\Rightarrow \frac{e^{-\lambda} \cdot \lambda^0}{0!} = \frac{e^{-\lambda} \cdot \lambda^1}{1!} \quad \text{i.e.,} \quad \lambda = 1.$$

Now $P(X=0) = \frac{e^{-\lambda} \cdot \lambda^0}{0!} = e^{-1} = 0.3678.$

ii By using recurrence formula, we have

$$P(x+1) = \frac{\lambda}{x+1} P(x).$$

$$\therefore P(x+1) = \frac{1}{x+1} \cdot P(x) \quad \because \lambda = 1. \quad \text{---(1)}$$

Now $P(1) = P(0) = 0.3678$

Put $x=1$ $P(2) = \frac{1}{2} \cdot P(1) = \frac{1}{2}(0.3678) = 0.1839$

Put $x=2$ in (1) $P(3) = \frac{1}{3} \cdot P(2) = \frac{1}{3}(0.1839) = 0.0613$

Put $x=3$ in (1)

$$P(4) = \frac{1}{4} P(3) = \frac{1}{4}(0.0613) = 0.015325$$

Put $x=4$ in (1)

$$P(5) = \frac{1}{5} P(4) = \frac{1}{5}(0.015325) = 0.003065$$

Prob: If a poisson distribution is such that $P(X=1) = \frac{3}{2} = P(X=3)$ find

i $P(X \geq 1)$ ii $P(X \leq 3)$ iii). $P(2 \leq x \leq 5)$.

Sol. Given $\frac{3}{2} P(X=1) = P(X=3)$.

i.e., $\frac{3}{2} \frac{e^{-\lambda} \cdot \lambda^1}{1!} = \frac{e^{-\lambda} \cdot \lambda^3}{3!}$ i.e., $\frac{3\lambda}{2} = \frac{\lambda^3}{6}$.

$\lambda^3 - 9\lambda = 0$ or $\lambda(\lambda^2 - 9) = 0$ $\lambda = 0, \pm 3$.

$\lambda = 3$. $\because \lambda > 0$.

Hence $P(X=x) = P(x) = \frac{e^{-3} \cdot 3^x}{x!}$

(i) $P(X \geq 1) = 1 - P(x < 1) = 1 - P(X=0) = 1 - \frac{e^{-3} \cdot 3^0}{0!}$

$= 1 - e^{-3} = 0.950213$.

(ii). $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$.

$= e^{-3}\left[\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!}\right]$

$= e^{-3}\left[1 + 3 + \frac{9}{2} + \frac{9}{2}\right) = 13e^{-3}$.

$= 0.6472318$.

(iii). $P(2 \leq x \leq 5) = P(X=2) + P(X=3) + P(X=4) + P(X=5)$.

$= e^{-3}\left[\frac{3^2}{2!} + \frac{3^3}{3!} + \frac{3^4}{4!} + \frac{3^5}{5!}\right]$.

$= 9e^{-3}\left[\frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{9}{40}\right]$

$= 9e^{-3}(1.6) = 0.7169337$.

Normal Distribution was discovered by Abraham B. Modi

It is also limiting case of B.D

* Total probability / Total Area of Normal Distribution.

We know that Total probability / Total area of N.D.

$$= \int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty} f(X=x)\,dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx.$$

which is the probability density function of N.D.

Total probability / Total area of N.D.

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

let $\frac{x-\mu}{\sigma} = z$

$x - \mu = \sigma z$

$dx = \sigma\, dz$.

$f(-x) = f(x)$ even fn

Integrand even fn.

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-z^2}{2}} dz$$

$$= \frac{1}{\sqrt{2\pi}} 2 \int_{0}^{\infty} e^{-z^2/2} dz$$

Put $z^2/2 = t$.

$z\,dz = dt$.

$$= \frac{1}{\sqrt{2\pi}} \cdot 2 \int_{0}^{\infty} e^{-t} \frac{dt}{\sqrt{2t}} dt.$$

$$= \frac{1}{\sqrt{\pi}} \int_{0}^{\infty} e^{-t} \cdot t^{-1/2} dt = \frac{1}{\sqrt{\pi}} \int_{0}^{\infty} e^{-1/2} \cdot t^{1/2-1} dt$$

$$= \frac{1}{\sqrt{\pi}} \cdot \Gamma 1/2 = \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi}.$$

$$= 1.$$

from Gramma fn.

$$\int_{0}^{\infty} e^{-x} x^{m-1} dx = \Gamma m.$$

$\Gamma 1/2 = \sqrt{\pi}$.

Total Area or total probability of N.D = 1.

Normal Distribution

It is the another limiting case of the binomial distribution for located (very Large) values of n, when neither p or q are very small.

A continuous random variable X is said to have a normal distribution, if its density function is given by

$$f(x=x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$-\infty < x < \infty$

$-\infty < \mu < \infty$

$\sigma > 0$.

Where μ = mean and σ = S.D. are two parameters of the distribution.

The random variable is called a normal random variable.

* Total probability / Total Area of normal Distribution is 1.

Mean of Normal Distribution

Consider the Normal Distribution with μ, σ as the Parametus $f(x; \mu\sigma) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$.

$$\text{Mean of ND} = E(x).$$

$$= \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

$$= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2}} dx.$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x \cdot e^{-\frac{(x-\mu)^2}{2}} dx.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z)\, e^{-z^2/2}\, dz.$$

Let $z = \frac{x-b}{\sigma}$, $dz = \frac{dx}{\sigma}$.

$x = z\sigma + b$.

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mu\, e^{-z^2/2}\, dz + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma z\, e^{-z^2/2}\, dz$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2}\, dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \cdot e^{-z^2/2}\, dz$$

$$= \frac{\mu}{\sqrt{2\pi}} \cdot \sqrt{2\pi} + \frac{\sigma}{\sqrt{2\pi}} \cdot 0$$

Mean of ND $= \underline{\mu}$.

Variance of ND

We know that $N.D = E(X^2) - [E(X)]^2$.

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\, dx.$$

$$= \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx.$$

Using substitution

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (\mu + \sigma z)^2 \cdot e^{-z^2/2} \sigma\, dz.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mu^2 e^{-z^2/2} dz + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 2\mu\sigma z\, e^{-z^2/2} dz + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 z^2 e^{-z^2/2} dz$$

$$= \frac{\mu^2}{\sqrt{2\pi}} \cdot \sqrt{2\pi} + 0 + \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2} dz$$

$$= \mu^2 + \frac{2\sigma^2}{\sqrt{2\pi}} \int_{0}^{\infty} z^2 e^{-z^2/2} dz$$

$$= \mu^2 + \frac{\sqrt{2}}{\sqrt{\pi}} \sigma^2 \int_{0}^{\infty} 2t\, e^{-t} \frac{dt}{\sqrt{2t}}.$$

$$= \mu^2 + \frac{2\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} t^{1/2} e^{-t} dt$$

$$= \mu^2 + \frac{2\sigma^2}{\sqrt{\pi}} \int_0^\infty t^{3/2-1} e^{-t}\, dt$$

$$= \mu^2 + \frac{2\sigma^2}{\sqrt{\pi}} \Gamma 3/2$$

$$= \mu^2 + \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \Gamma 1/2. \quad = \mu^2 + \frac{\sigma^2}{\sqrt{\pi}} \cdot \sqrt{\pi}.$$

$E(X^2) = \mu^2 + \sigma^2$

Variance of N.D $= E(X^2) - [E(X)]^2$

$= \mu^2 + \sigma^2 - \mu^2 = \sigma^2.$

Mode of N.D.

We know that Mode is the value of x at which the probability $f(x) - f(x)$ has maximum.

Here $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{\frac{(x-\mu)^2}{2}} \quad -\infty < x < \infty$

Taking log on both sides.

$$\log f(x) = \log \left[\frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \right]$$

$$= \log \frac{1}{\sqrt{2\pi}\sigma} + \left(-\frac{1}{2} \left[\frac{x-\mu}{\sigma}\right]^2 \right).$$

diff wrt x

$$\frac{1}{f(x)} \cdot f'(x) = 0 - 2 \cdot \frac{(x-\mu)}{2\sigma^2}$$

$$f'(x) = -\frac{(x-\mu)}{\sigma^2} f(x) \quad \text{---} (2).$$

Again diff. wrt x

$$f''(x) = -\frac{1}{\sigma^2} \left[1 \cdot f(x) + (x-\mu) \cdot f'(x) \right]. \quad \text{---} (2)$$

Sub. (2) in (1)

$$f''(x) = -\frac{1}{\sigma^2}\left[f(x) + (x-\mu)\left(-\frac{(x-\mu)}{\sigma^2}\cdot f(x)\right)\right]$$

$$= -\frac{f(x)}{\sigma^2}\left[1 - \left(\frac{x-\mu}{\sigma}\right)^2\right].$$

let $f'(x) = 0$

$$-\frac{(x-\mu)}{\sigma^2}\cdot f(x)$$ here $f(x) \neq 0$.

$\therefore -\frac{(x-\mu)}{\sigma^2} = 0 \Rightarrow x = \mu$.

Sub. in (2)

$$f''(x) = -\frac{f(x)}{\sigma^2}(1-0). = -\frac{f(x)}{\sigma^2} = -ve$$

σ^2 always +ve
$f(x)$ " "

$\therefore$ at $x = \mu$ $f'(a) = 0$

$f''(a) < 0$

hence $f(x)$ has max. at $x = \mu$.

Mode of N.D is $x = \mu$,

In the N.D. Mean, Median, Mode are coincide
i.e, mean = median = mode.

Standard Normal Variant (SNV).

If X is a ~~random~~ Continuous random variable the following ND with mean μ and variance σ^2 then the SNV is denoted by Z.

$$Z = \frac{x-\mu}{\sigma} = \frac{x - E(x)}{\sqrt{V(x)}}.$$

mean of SNV.

$$= \text{mean of } Z.$$

$$= E(Z)$$

$$= E\left(\frac{x-\mu}{\sigma}\right)$$

$$= \frac{1}{\sigma} E(x-\mu).$$

$$= \frac{1}{\sigma}[E(x) - E(\mu)]$$

$$= \frac{1}{\sigma}[\mu - \mu] = 0.$$

$E(c) = c$

$E(cx) = cE(x)$

$E(X+Y) = E(X) + E(Y)$

Mean of SNV = 0

Variance of SNV.

Variance of SNV = Var (z)

$$= \text{Var}\left(\frac{x-\mu}{\sigma}\right)$$

$$= \frac{1}{\sigma^2} \text{Var}(x-\mu)$$

$$= \frac{1}{\sigma^2} \text{Var}(x) = \frac{1}{\sigma^2} \cdot \sigma^2$$

$$= 1.$$

$\therefore$ Variance of SNV = 1

$\text{Var}(c) = 0$

$\text{Var}(cx) = c^2 \text{Var}(x)$

$\text{Var}(ax+b) = a^2 \text{Var}(x)$.

The standard normal variant probability curve is given by.

a, b are numbers.

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad -\infty < z < \infty.$$

Variance of SNV:

$\text{Var}(c) = 0$

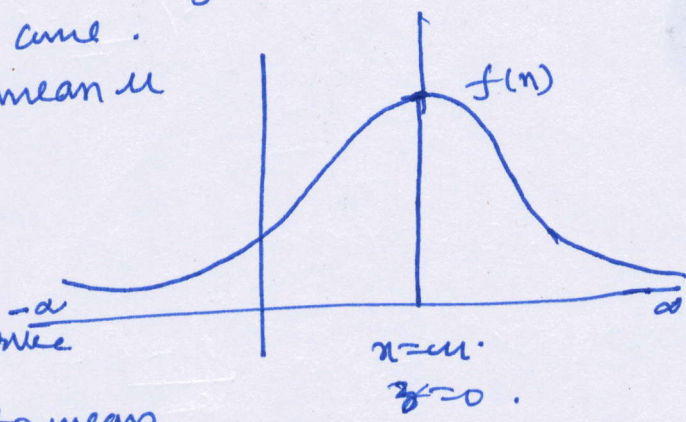
$\text{Var}\, c(x) = c^2 \text{Var}(x)$.

$\text{Var}(ax+b) = a^2 \text{Var}(x)$.

Properties of Normal Distribution Curve

1. The graph of normal distribution $y = f(x)$ in the XY-plane is known as normal curve.
The normal distribution curve with the mean μ and S.D σ is given by

$$f(x=x) = f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



2. The curve is a bell shaped curve and Symetrical with respect to mean.
i.e., about the line $x = \mu$. and the two tails on the right and left sides of mean (μ) tends to infinity.
The top of the bell is directly above the mean μ.

3. The curve is Symetrical about the line $x = \mu$ or $z = 0$

4. Since the distribution is Symetrical mean, median and mode coincide i.e, mean = median = mode = μ.

5. Since Mean = Median = μ, the ordinate at $x = \mu$ or $z = 0$ divides the whole area into two equal parts, further the total area under the prob. curve is unity. Therefore the area to the right of the ordinate as well as left of the ordinate at $x = \mu$ or $z = 0$ is 0.5.

6. Area under the N.D curve:
One of the most fundamental property of the Normal probability curve is the area property.
* The total area under the normal probability curve is unity. * The area bounded by the curve f(x)

X-axis and the ordinates $x=a$ and $x=b$ is given by the ordinate integral.

$$\int_a^b f(x)\,dx.$$

i.e., $$P(a \le x \le b) = \int_a^b f(x)\,dx.$$

* Let us compute the area or probability of $(\mu < x < a)$ as follows $P(\mu \le x \le a) = \int_{\mu}^{a} f(x)\,dx.$

$$= \int_{\mu}^{a} \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

put $z = \frac{x-\mu}{\sigma}$.

$dx = \sigma\, dz$.

$x = \mu \Rightarrow z = 0$

$x = a \Rightarrow z = \frac{a-\mu}{\sigma} = z_1$, say

$$= \int_0^{z_1} \frac{1}{\sqrt{2\pi}} \cdot e^{-z^2/2}\, dz.$$

$$\therefore P(\mu < x < a) = \int_0^{z_1} \frac{1}{\sqrt{\pi}} e^{-z^2/2}\, dz.$$

This integral is called probability integral. This definite integral which gives the area under the normal ~~distribution~~ probability curve bounded by X-axis and ordinate of $x=\mu$ and $x=a$ (or) $z=0$ and $z = \frac{x-\mu}{\sigma} = z_1$ have been evaluated and the tables are available for different values of z_1 at intervals of 0.01

In particular $x = \mu - \sigma$, $z = -1$, $x = \mu + \sigma$.

$$P(\mu - \sigma < x < \mu + \sigma) = \phi(-1 < z < 1) = \phi(-1 < z < 0) + \phi(0 < z < 1).$$

$$= 2 * \phi(0 < z < 1)$$

due to Symmetry

$$P(\mu - \sigma < x < \mu + \sigma) = 2 * \phi(0 < z < 1)$$

$$= 2 * 0.3413.$$

$P(\mu-\sigma < x < \mu+\sigma) = 0.6826.$

Illy $P(\mu-2\sigma < x < \mu+2\sigma) = 2 * \phi(0 < z < 2)$

$= 2 * 0.4772$

$= 0.9544$

Illy. $P(\mu-3\sigma < x < \mu+3\sigma) = 2 * (0 < z < 3).$

$= 2 * \phi(0 < z < 3). = 0.9974 \cong 1.$

∴ The area under the normal probability curve with ordinates at $x = \mu-3\sigma$ and $x = \mu+3\sigma$ is 0.997% & approx. equal to 1.

Hence for all practical purpose the range of normal distribution is 6σ $(3\sigma+3\sigma)$.

* X-axis is the asymptote to the curve.

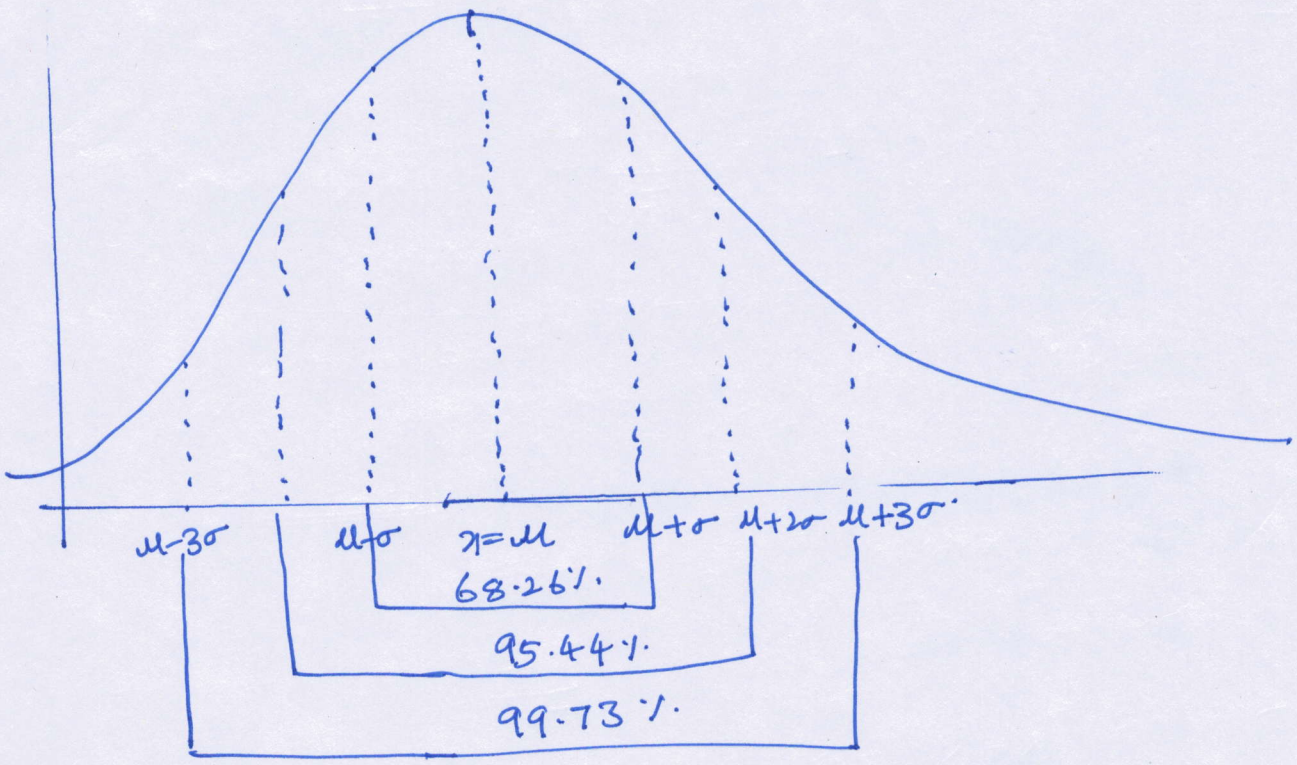
* Linear Combination of independent normal variates is also normal variate

* Area under the normal curve is distributed as follows.

(i) Area of normal curve between $\mu-\sigma$ and $\mu+\sigma$ is 68.26%.

(ii) Area of normal curve between $\mu-2\sigma$ and $\mu+2\sigma$ is 95.44%.

(iii) Area of normal curve between $\mu-3\sigma$ and $\mu+3\sigma$ is 99.7%.



We may follow the following steps to obtain the Sol. of N.D in a Systamatic manner.

Step1. If the values are in X, the Convert into Z.

Step2. Reognise these values of z in the bell shaped curve

step3. Use the Normal Distribution table to find the area corresponding to the z values of the previous step.

Step4: Then simplify, we get the required answer.

Obs:

1. $P(x > \mu) = P(z > 0) = 0.5$.

2. $P(x > a) = P(z > z_1)$. $\quad z_1 = \frac{a-\mu}{\sigma}$.

$= P(z > 0) - P(0 > z > z_1)$.

$= 0.5 - P(0 < z < z_1)$.

Prob: Find the area under the Normal probability curve.

a. To the left of $z = -1.78$

b. To the left of $z = 0.56$

c. To the right of $z = -1.45$

d Corresponding to $z \geq 2.16$

e corresponding to $-0.80 \leq z \leq 1.53$.

f. to the left of $z = -2.52$ and to the right of $z = 1.83$

g. Corresponding to $|z| \geq 1$.

Sol.

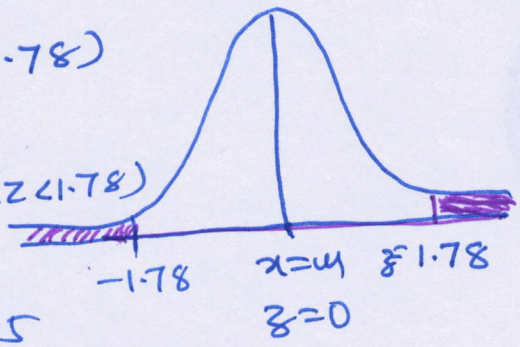
(a). $f(z \leq -1.78) = f(z \geq 1.78)$

$= f(z > 0) - f(0 < z < 1.78)$

$= 0.5 - 0.4625$

from table.

$\therefore f(z \leq -1.78) = 0.0375$

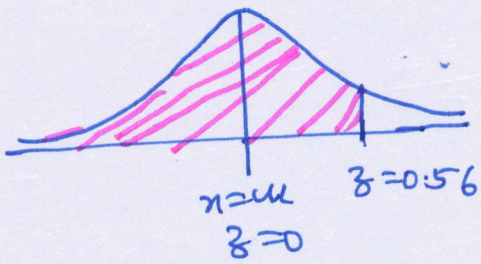


(b). The left of $z = 0.56$.

$P(z < 0.56) = f(z < 0) + f(z \leq 0.56)$

$= 0.5 + 0.2123$

$= 0.7123.$



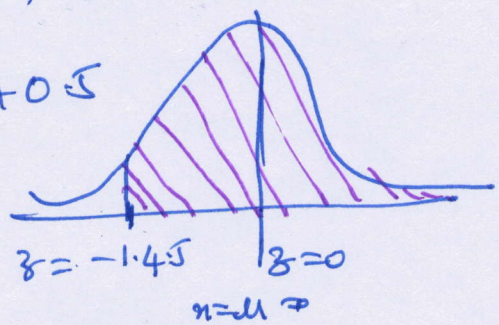
c. The right of $z = -1.45$.

$f(z \geq -1.45) = f(0 > z > -1.45) + f(z > 0).$

$= f(0 < z < -1.45) + 0.5$

$= 0.4265 + 0.5$

$f(z \geq 1.45) = 0.9265.$

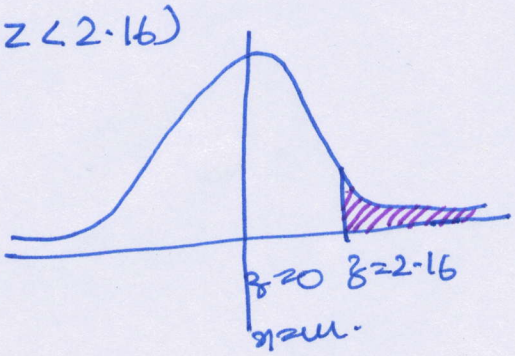


d. Corresponding to $z \geq 2.16$.

$f(z \geq 2.16) = f(z \geq 0) - f(0 < z < 2.16)$

$= 0.5 - 0.4846$

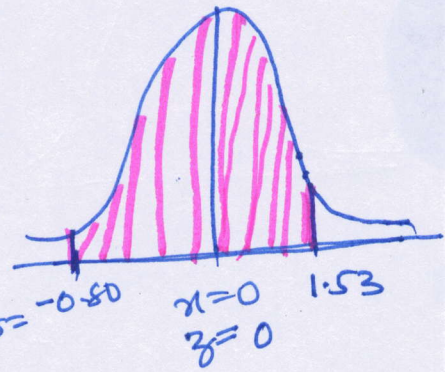
$= 0.0154.$



e. Corresponding to $-0.80 \le Z \le 1.53$.

$f(-0.80 \le z \le 1.53)$

$= f(-0.8 \le z \le 0) +$
$f(0 \le z \le 1.53)$.



$= f(0 \le z \le 0.8) + f(0 \le z \le 1.53)$.

$= 0.2881 + 0.4370$

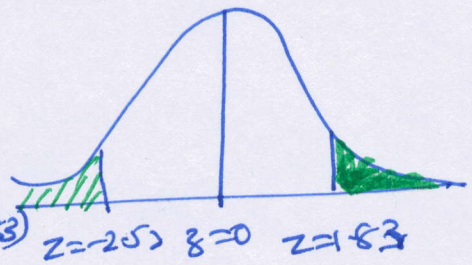
$= 0.7251$.

f. to the left of $z = -2.52$ and to the right of $z = 1.83$.

$P(z \le -2.52) + P(z \ge 1.83)$.

$= [P(z > 0) - f(0 < z < 2.52)] +$

$[P(z > 0) - f(0 < z < 1.83)]$



$= 0.5 - 0.49414 + 0.5 - 0.4664$.

$= 0.0395$.

(g). corresponding $|z| \ge 1$.

$|z| \ge 1$. ; $\pm z \ge 1$. $-z \ge 1$. & $z \ge 1$.

$z \le -1$ & $z \ge 1$.

$P(z \le -1) + f(z \ge 1)$.

$= 1 - [f(-1 \le z \le 1)]$

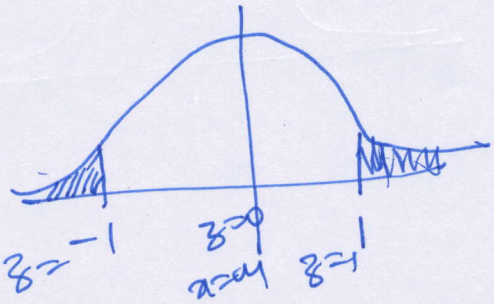
$= 1 - [f(-1 \le z \le 0) + f(0 \le z \le 1)]$

$= 1 - f(0 \le z \le 1) + P(0 \le z \le 1)$

$= 1 - 2 f(0 \le z \le 1)$.

$= 1 - 2 \times 0.3413$

$= 0.3174$ //.



Prob: For a normally distributed variati with mean 1 and S.D 3. find the probability that $-1.43 \le x \le 6.19$.

Given $\mu = 1$, $\sigma = 3$.

When $x = -1.43$ $\quad z = \frac{x-\mu}{\sigma} = \frac{-1.43-1}{3} = -0.81$

~~z say.~~

When $x = 6.19$ $\quad z = \frac{6.19-1}{3} = \frac{5.19}{3} = 1.73$.

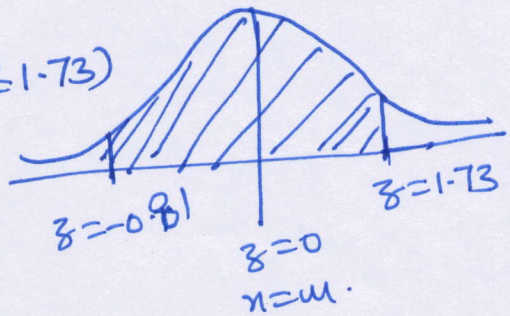
$f(-1.43 \le x \le 6.19) = f(-0.81 \le z \le 1.73)$.

$= f(-0.81 \le z \le 0) + f(z \le 1.73)$

$= f(0 \le z \le 0.81) + f(z \le 1.73)$

$= 0.2910 + 0.4582$

$= 0.7492$.



Prob. The marks obtained in mathematics 1000 students is normally distributed with mean 78% and S.D 11%. Determine.

(i) How many students got marks above 90%.

Sol:

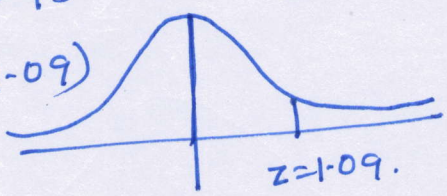
Given mean $= \mu = 0.78$ and $SD = 0.11 \Rightarrow \sigma = 0.11$.

When $x = 0.9$ $z = \frac{x - \mu}{\sigma} = \frac{0.9 - 0.78}{0.11}$.

$z = 1.09$.

Hence the number of students with marks more than 90% is $f(z > 1.09)$.

$= f(z > 0) - f(0 < z < 1.09)$



$= 0.5 - 0.3621$

$= 0.1379$

Hence the number of students with marks more than 90% $= 0.1379 \times 1000 = 137.9 \simeq 138$.

12. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that (*i*) exactly two will be defective (*ii*) at least two will be defective (*iii*) none will be defective.

[JNTU (K) Nov. 2012 (Set No. 3)]

13. If the chance that one of the ten telephone lines is busy at an instant is 0.2,
    (*i*) What is the chance that 5 of the lines are busy ?
    (*ii*) What is the most probable number of busy lines and what is the probability of this number ?
    (*iii*) What is the probability that all the lines are busy ?

14. A box contains 100 transistors, 20 of which are defective and 10 are selected at random, find the probability that
    (*i*) all are defective
    (*ii*) at least one is defective
    (*iii*) all are good
    (*iv*) at most 3 are defective.

15. In a certain city only 50% of the students are capable of doing college work actually go to college. Assuming that this claim is true, find the probability that among 18 such a capable students
    (*i*) exactly 10 will go to college
    (*ii*) at least 2 will go to college
    (*iii*) at most 17 will go to college.

16. The probability of a newly generated virus attacked to the computer system will corrupt the 4 files out of 20 files which are opened in an hour. If 20 files are opened in an hour, find the probability that
    (*i*) at least 10 files are corrupted
    (*ii*) exactly 3 files are corrupted
    (*iii*) all files are corrupted
    (*iv*) all files are safe
    (*v*) more than 2 but not more than 5 files are corrupted.

17. The overall percentage of failures in a certain examination is 20. If six candidates appear in the examination, what is the probability that at least five pass the examination ?

18. A pair of dice is thrown. Find the probability of getting a sum 11 (*i*) once (*ii*) twice.

19. Find the probability that in tossing a fair coin 6 times there will appear (*i*) 0 (*ii*) 1 (*iii*) 2(*iv*) 3 (*v*) 4 (*vi*) 5 and (*vii*) 6 heads.

20. 5 coins are tossed at a time, 3200 times. Find the frequencies of the distribution of heads and tails and tabulate the results. Also determine the mean number of success and standard deviation.

21. Fit a Binomial Distribution to the following data

x	0	1	2	3	4	5
f	38	144	342	287	164	25

[JNTU 2004S (Set No.2)]

22. The following data show the number of seeds germinating out of 10 on damp filter for 80 set of seeds. Fit a Binomial Distribution to the data.

x	0	1	2	3	4	5	6	7	8	9	10
f	6	20	28	12	8	6	0	0	0	0	0

[JNTU (K) May 2012 (Set No. 4)]

23. 7 coins are tossed at a time, 128 times. The number of heads observed at each throw is recorded and the results are given below :

No. of heads	0	1	2	3	4	5	6	7
Frequency	7	6	19	35	30	23	7	1

Fit a binomial distribution to the data assuming that the coins are (*i*) biased (*ii*) unbiased ?

[JNTU 2000 S]

24. Fit a Binomial Distribution to the following frequency data.

x	0	1	2	3	4
f	28	62	46	10	4

[JNTU Jan. 2007]

25. In 800 families with 5 children each, how many families would be expected to have (*i*) 3 boys and 2 girls (*ii*) 2 boys and 3 girls (*iii*) no girl (*iv*) at the most two girls.

[JNTU Nov. 2012 (Set No. 4)]

26. In a binomial distribution consisting of 5 independent trails, probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter 'p' of the distribution.

[JNTU (A) June 2018]

ANSWERS

1. 0.6243 **2.** 0.2461 **3.** $\frac{3}{16}$ **4.** 0.9822 **6.** 6 **7.** 0.1937

8. (*i*) 0.3487 (*ii*) 0.3874 (*iii*) 0.9298 **9.** (*i*) 0.109375 (*ii*) 0.9648

10. (*i*) 0.890625 (*ii*) 0.65625 **11.** (*i*) 0.26 (*ii*) 0.0000244 (*iii*) 0.7674

12. (*i*) 0.2301 (*ii*) 0.3412 (*iii*) 0.2833

13. (*i*) 0.02642 (*ii*) 2, 0.0457 (*iii*) 1.023×10^{-7}

14. (*i*) $\frac{1}{5^{10}}$ (*ii*) 0.8926 (*iii*) 0.1074 (*iv*) 0.879128

15. (*i*) 0.1669 (*ii*) 0.9999 (*iii*) 0.999999

16. (*i*) 0.000004526 (*ii*) 0.00107374 (*iii*) $\frac{1}{5^{12}}$ (*iv*) 0.999999 (*v*) 0.422249

17. 0.65536 **18.** (*i*) 0.105 (*ii*) 0.00308 **19.** $\frac{1}{2^6}, \frac{6}{2^6}, \frac{15}{2^6}, \frac{20}{2^6}, \frac{15}{2^6}, \frac{6}{2^6}, \frac{1}{2^6}$

20.

x	0	1	2	3	4	5
$p(x)$	100	500	1000	1000	500	100

; $\mu = 2.5$, $\sigma = 1.118$

21. 33, 162, 316, 309, 151, 29 **22.** $80\,[0.8725 + 0.2175]^{10}$

23. (*i*) $128\,[0.517 + 0.483]^7$ (*ii*) 1, 7, 21, 35, 35, 21, 7, 1

EXERCISE 8.3

1. On an average, 2 vehicles pass by a road per minute. Find the probability of 0, 1, 2, 3, 4, 5 vehicles per minute.
2. The fraction defective in a bulk is 4%. Determine the probability of not more than one defective in a sample of size 10.
3. If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs, there are 5 defective bulbs.
4. Find the probability of finding at most 5 defective fuses in a box of 200 fuses if the experience shows that 2% of such fuses will be found defective.
5. Suppose that a book of 200 pages contains 20 printing mistakes. Assume that these errors are randomly distributed throughout the book and x, the number of errors per page has a poisson distribution. Find the probability that 30 pages selected at random will be free of errors.
6. A manufacturer known that the razor blades he makes contain on an average 0.5% of defectives. He packs them in packets of 5. What is the probability that a packet picked at random will contain 3 or more faulty blades.
7. Suppose 3% of bolts made by a machine are defective, the defects occurring at random during production. If bolts are packed 50 per box, find the probability that a given box will contain 5 defectives.
8. Assume that the probability of an individual coal miner being killed in a mine accident during a year is $\frac{1}{2400}$. What is the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year.
9. In a city, 6% of all drivers get at least one parking ticket per year. Determine the probabilities that among 80 drivers
   (*i*) 4 drivers will get at least one parking ticket in a year.
   (*ii*) at least 3 drivers will get at least one parking ticket in a year.
10. Using Poisson distribution find the probability that the ace of spades will be drawn from a pack of well shuffled cards at least once in 104 consecutive trials.
11. The number of arrivals of customers during any day follows Poisson distribution with mean of 5. What is the probability that the total number of customers on two days selected at random is less than 2 ?
12. An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year ?
13. What is the probability that an ace of spade will not be drawn even once from a deck of cards in 100 consecutive trials.
14. Out of 100 balls 50 are red and the rest white. If 60 balls are picked up at random, what is the probability of picking up (*i*) 3 red balls (*ii*) not more than 3 red balls in the sample.

15. A person receives an average of 4 telephone calls a day. Assuming Poisson distribution what is the probability that on a given day he will receive (*i*) no telephone call (*ii*) exactly only call (*iii*) at least two telephone calls

16. The random variable X has a Poisson distribution. If $P(X = 1) = 0.01487$ and $P(X = 2) = 0.04461$, then find $P(X = 3)$.

17. If X is a Poisson variate such that $P(X = 0) = P(X = 2) + 3P(X = 4)$, find (*i*) the mean of X (*ii*) $P(X \leq 2)$

[JNTU 2004S (Set No. 3)]

18. Using recurrence formula find the probabilities when $x = 0, 1, 2, 3, 4$ and 5, if the mean of Poisson distribution is 3.

[JNTU 2005 (Set No. 4)]

19. Suppose that X has a Poisson distribution. If $P(X = 2) = \frac{2}{3} P(X = 1)$, find (*i*) $P(X = 0)$ and (*ii*) $P(X = 3)$.

20. If X and Y are independent Poisson variates such that $P(X = 1) = P(X = 2)$ and $P(Y = 2) = P(Y = 3)$. Find the variance of (*i*) X (*ii*) Y (iii) $X - 2Y$.

21. If X follows a poisson law such that $2P(X = 0) = P(X = 2)$, find (*i*) $p(X \leq 3)$ (*ii*) $p(2 < X \leq 5)$ (*iii*) $p(X \geq 3)$.

[JNTU (A) Nov. 2011(Set No. 2)]

22. If X is a poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X.

[JNTU (K) May 2012 (Set No. 1)]

23. If X is a poisson variate such that $P(X = 2) = 45 P(X = 6) - 3 . P(X = 4)$, find (*i*) $P(X \geq 1)$ (*ii*) $P(X < 2)$.

24. A Poisson distribution has a double mode at $x = 1$ and $x = 2$. What is the probability that x will have one or the other of these two values ?

[JNTU (H) May 2012(Set No. 4)]

25. If X is a Poisson variate such that $P(X = 0) = P(X = 1) = k$, determine k.

26. Fit a Poisson distribution for the following data and calculate the expected frequencies.

x	0	1	2	3	4	5	6	7	8
f	56	156	132	92	37	22	4	0	1

27. The distribution of typing mistakes committed by a typist is given below assuming the distribution to be Poisson, find the expected frequencies

x	0	1	2	3	4	5
$f(x)$	125	95	49	20	8	3

ANSWERS

1. 0.1353, 0.2706, 0.2706, 0.1804, 0.0902, 0.0361

2. 0.9384 **3.** 0.1008 **4.** 0.7845

5. 0.05 **6.** 0.000002555 **7.** 0.01412

8. 0.08 **9.** (i) 4.8 (ii) 0.8575 **10.** 0.865

11. 4.994×10^{-4} **12.** 0.9998 **13.** 0.15

14. (*i*) 0.2241 (*ii*) 0.5976 **16.** 0.08922 **17.** (*i*) 1.21 (*ii*) 0.8773

18. 0.147, 0.2205, 0.2205, 0.1653, 0.099, 0.0495

19. (*i*) 0.264 **20.** (*i*) 2 (*ii*) 3 (*iii*) 4

21. (*i*) 0.0861 (*ii*) 0.0308 (*iii*) 0.32

22. 1, 1 **23.** (*i*) 0.864 (*ii*) 0.408 **24.** 0.542

25. $1/e$ **26.** 70, 137, 135, 89, 44, 17, 6, 20

27. 110, 110, 55, 18, 5, 1 **28.** 301, 361, 217, 87, 26, 6, 1, 0

Normal Distribution Assignment

1. The mean and standard deviation of a normal variate are 8 and 4 respectively, find
(*i*) $P(5 \le x \le 10)$ (*ii*) $P(x \ge 5)$ **[JNTU 2005 S, (H) Nov. 2009 (Set No. 1)]**
(*iii*) $P(10 \le x \le 15)$ (*iv*) $P(x \le 15)$ **[JNTU 2004 (Set No.2)]**

2. If the mean and standard deviation of a normal distribution are 70 and 16, find $P(38 < x < 46)$. **[JNTU 2005S (Set No.4)]**

3. If X is normally distributed with mean 70 and standard deviation 16, find
(*i*) $P(38 \le X \le 46)$ (*ii*) $P(82 \le X \le 94)$ (*iii*) $P(62 \le X \le 86)$

4. Find the probability that a random variable having standard normal distribution will take on a value between 0.87 and 1.28. **[JNTU 2001]**

5. If X is a normally distributed with mean 30 and standard deviation 5, find $P(|X-30|>5)$.

6. Find the mean and standard deviation of a normal distribution in which 31% of items are under 45 and 8% are over 64. [**Hint :** Refer Example 4]

[JNTU 2003, (A) Dec 2009 (Set No.2), (K) Nov. 2012, Dec. 2014 (Set No. 2)]

7. (*i*) The mean height of 500 students is 151 cm. and the standard deviation is 15 cm. Assuming that the heights are normally distributed, find how many students heights lie between 120 and 155 cm.

(*ii*) The heights of 1000 students are normally distributed with a mean of 174.5 cm and a standard deviation of 6.9 cm. Assuming that the heights are recorded to the nearest half-cm, how many of these students would you expect to have heights :

(*a*) Less than 160.0 cms ? (*b*) Between 171.5 and 182.0 cms inclusive ?

(*c*) Greater than or equal to 188.0 cms. **[JNTU (A) June 2018]**

8. The life of electronic tubes of a certain type may be assumed to be normally distributed with mean 155 hours and standard deviation 19 hours. Determine the probability that the life of a randomly chosen tube

(*i*) is between 136 hours and 174 hours.

(*ii*) less than 117 hours (*iii*) will be more than 395 hours.

9. In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. Find the number of bulbs likely to burn for

(*i*) more than 2150 hours (*ii*) less than 1950 hours and

(*iii*) more than 1920 hours and but less than 2160 hours.

10. Estimate the minimum marks a student must get in order to receive an 'A' grade if the top 10% of the students are awarded 'A' grade in an examination where the mean of the marks is 72 and standard deviation is 9.

11. When the mean of marks was 50% and standard deviation 5% then 60% of the students failed in an examination. Determine the 'grace' marks to be awarded in order to show that 70% of the students passed. Assume that the marks are normally distributed, **[JNTU (A) May 2012 (Set No. 1)]**

12. The mean and standard deviation of marks scored by a group of students in an examination are 15 and 1.5 respectively. If only 25% of the students are promoted, determine the minimum marks required for the promotion.

13. Find the probability of getting at least 55 heads when 100 coins are tossed using normal distribution.

14. Twelve coins are tossed together. Find the probability of getting 8,9 or 10 heads in a single toss using normal distribution.

15. Ten cards are drawn from a deck of 52 cards. Find the probability of getting 2 to 5 diamonds using normal distribution.

16. In a referendum 60% of voters voted in favour. A random sample of 200 voters was selected. Find the probability by using normal distribution that in the sample

(*i*) more that 130 voted in favour ?

(*ii*) between 105 and 130 inclusive voted in favour ?

(*iii*) 120 voted in favour ?

17. Determine the probability of getting at least 60 heads when 100 coins are tossed.

18. The weekly wages of 1000 workers are normally distributed around a mean of Rs. 70 and S.D. of Rs. 5/-. Estimate the number of workers whose weekly wages will be

(*i*) between Rs. 70 and Rs. 72

(*ii*) between Rs. 69 and Rs. 72. **[JNTU 2007S (Set No.1)]**

ANSWERS

1. (*i*) 0.465 (*ii*) 0.7734 (*iii*) 0.6514 (*iv*) 0.0401

2. 0.044

3. (*i*) 0.044 (*ii*) 0.1598 (*iii*) 0.5328

4. 0.0919

5. 0.3174

6. $\mu = 50, \sigma = 10$

7. 294

8. (*i*) 0.6826 (*ii*) 0.0228 (*iii*) 0

9. (*i*) 67 (ii) 184 (*iii*) 1909

10. 84

11. 3.85

12. 28

13. 0.1841

14. 0.1887

15. 0.7530

16. (*i*) 0.0648 (*ii*) 0.8964 (*iii*) 0.0575

17. 0.02872

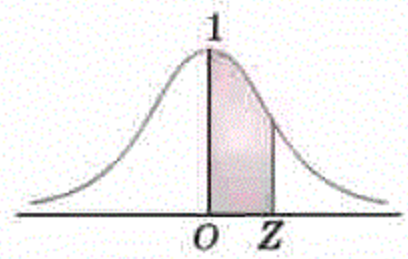
18. 155, 235

20. 1, 4, 6, 4, 1

21. 2, 1, 3, 36, 16, 3, 0

22. 1, 4, 6, 4, 1

Table III : Area under the Normal curve



x	0	1	2	3	4	5	6	7	8	9
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0754
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2258	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2996	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993